

Biochemistry of CVS For Pc-Medical Students

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Lipoprotein Metabolism and Hyperlipidemia

Learning Objectives

- By the end of this lecture, the students will be able to:
 - Define and list the types, structure and composition of lipoproteins
 - Understand various functions of lipoprotein particles
 - Compare the functions of lipoprotein particles and their implications in disease
 - Understand the metabolism of chylomicrons, VLDL and LDL particles
 - Discuss the functions of lipoprotein lipase and its role in disease
 - List the diseases due to imbalance in the metabolism of lipoproteins

Plasma lipoproteins

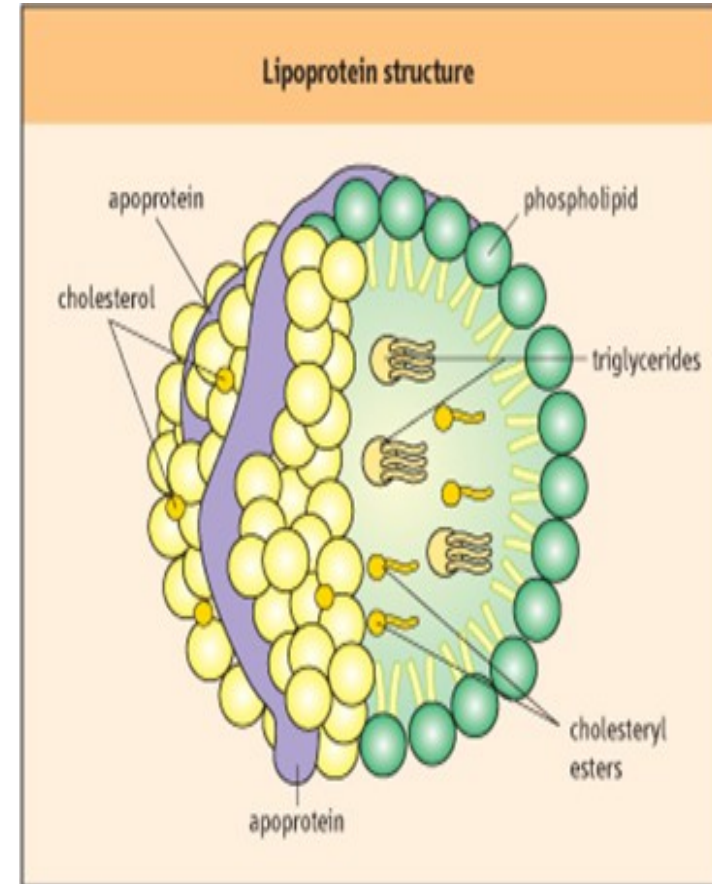
- ❖ Molecular **complexes of lipids** and **specific proteins**.
- ❖ Dynamic particles
 - constant state of
 - »**synthesis**,
 - »**degradation** and
 - »**removal** from the plasma.
- ❖ Lipoprotein particles include:
 - 1.Chylomicrons.**
 - 2.Very low density lipoproteins (VLDL)**
 - 3.Low density lipoproteins (LDL).**
 - 4.High density lipoproteins (HDL).**

Plasma lipoproteins

- ❖ Lipoproteins function to
 - keep **lipids soluble** as
 - » they transport them in plasma, and
 - provide an **efficient mechanism** for
 - » delivering **their lipid contents** to the tissues
- ❖ The **principal lipids** carried by **lipoprotein particles** are
 - **TRIACYLGLYCEROLS** and
 - **CHOLESTEROL (free or esterified)** .

Plasma lipoproteins

- ❖ lipoproteins are composed of
 - a neutral lipid core containing triacylglycerol and
 - cholesteryl esters or both,
 - surrounded by **apolipoproteins** (apoproteins)
 - phospholipid and
 - non-esterified cholesterol



Lipoprotein composition

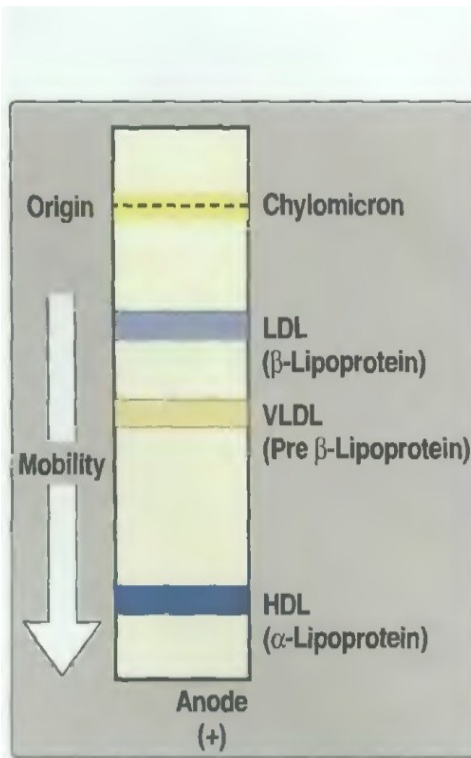


Figure 18.15
Electrophoretic mobility of plasma lipoproteins.

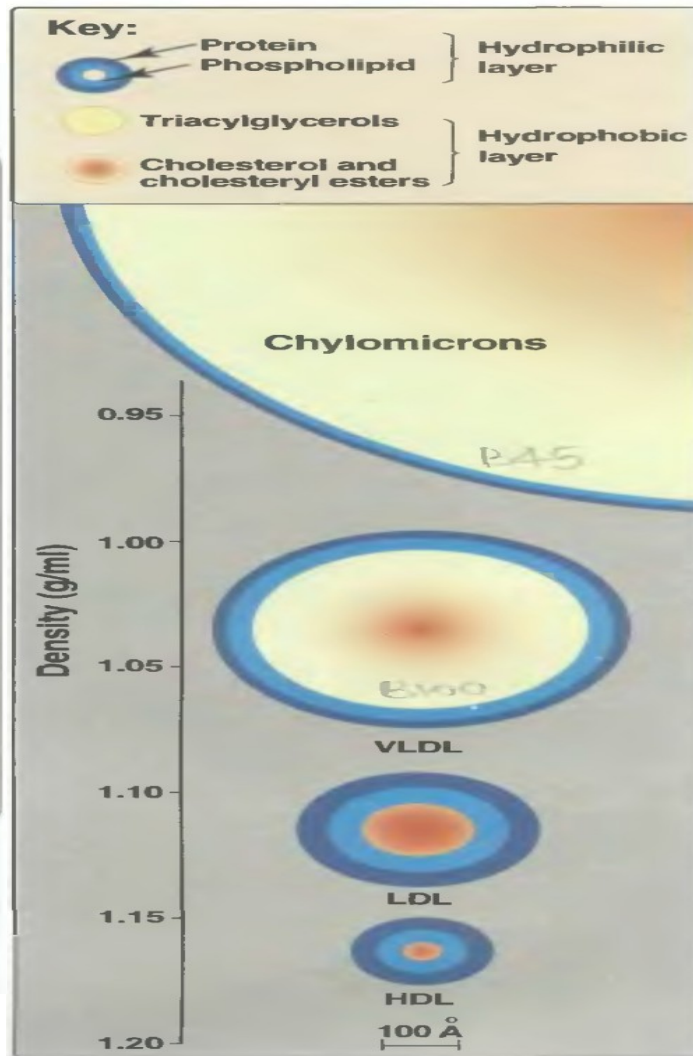


Figure 18.13
Approximate size and density of serum lipoproteins. Each family of lipoproteins exhibits a range of sizes and densities; this figure shows typical values. [Note: The width of the rings approximates the amount of each component.]

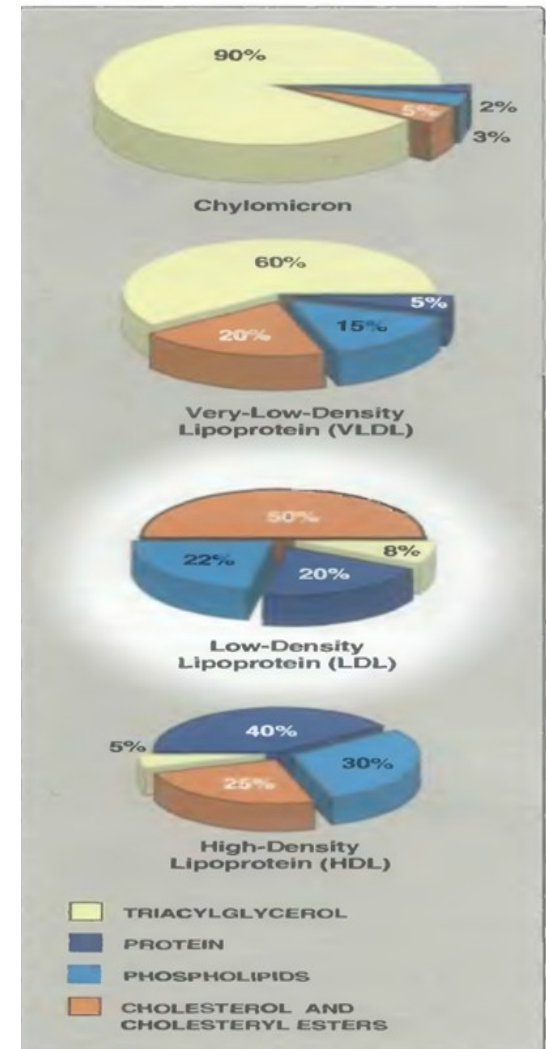


Figure 18.19
Composition of the plasma lipoproteins. Note high concentration of cholesterol and cholesteryl esters in LDL.

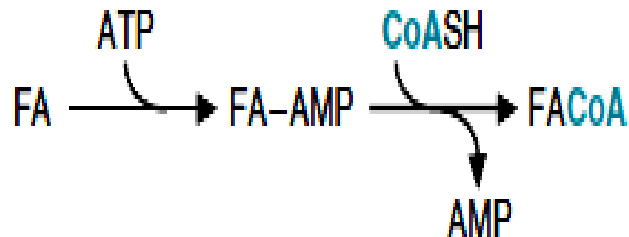
Apolipoproteins (Apoproteins)

- ❖ The apo-lipoproteins functions including:
 - Serving as **structural components** of the particles.
 - Providing **recognition sites for cell – surface receptors.**
 - Serving as **activators** in lipoprotein metabolism.
- ❖ Apolipoproteins are divided by
 - **structure** and **function** into classes A to H
 - ✓ with most classes having sub-classes for example, apo A- I and apo C- II.

Synthesis of Chylomicrons

- ▶ Within the intestinal **epithelial cells**, **the FAs & 2-MGs** are
 - **Re-esterfied** in the smooth ER to form TAGs
- ▶ The **FAs** are activated to **fatty acyl CoA** by
 - **Fatty acyl coA synthetase(Thiokinase)**
- ▶ A **fatty acyl CoA** then reacts with a 2-MG to form a DAG,
 - w/c reacts with another **fatty acyl CoA** to form a **Tag-D5**
- ▶ The rxn's for **TAG synthesis** in intestinal cells differ from
 - those in liver & adipose cells in that;
 - **2-MG is an intermediate** in TAG in **intestinal cells**
 - **Phosphatidic acid** is the intermediate in other tissues

Activation of fatty acids



Triacylglycerol synthesis

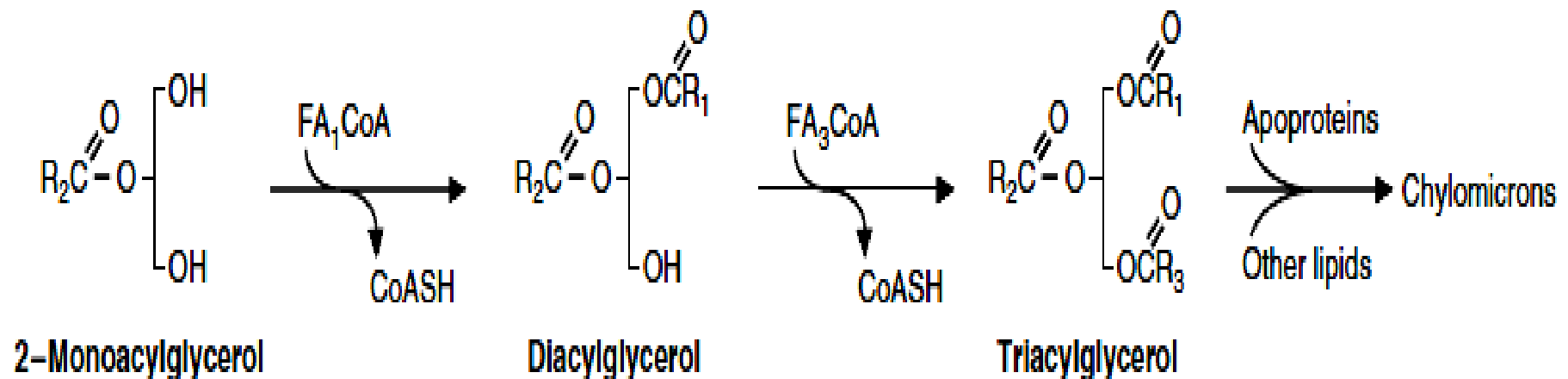


Fig-D5. Resynthesis of TAGs in Intestinal Epithelial Cells

Synthesis of...(Cont'd)

❖ Intestinal cells

- package TAGs together with
 - **proteins & phospholipids** in chylomicrons (CM)
- w/c are lipoprotein particles that do not
 - **readily coalesce** in aqueous solutions  [Fig-D6](#)
- Chylomicrons also contain **cholesterol & fat-soluble vitamins**
- The major **apoprotein** associated with CMs
 - as they leave the intestinal cells is **apoB-48**

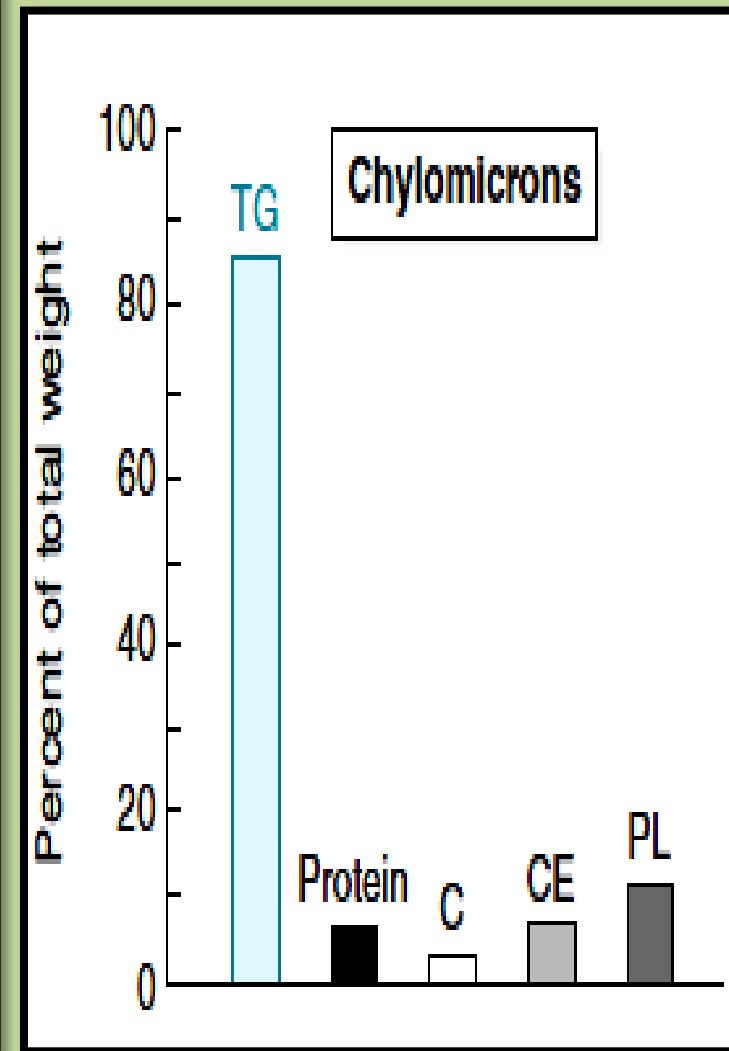
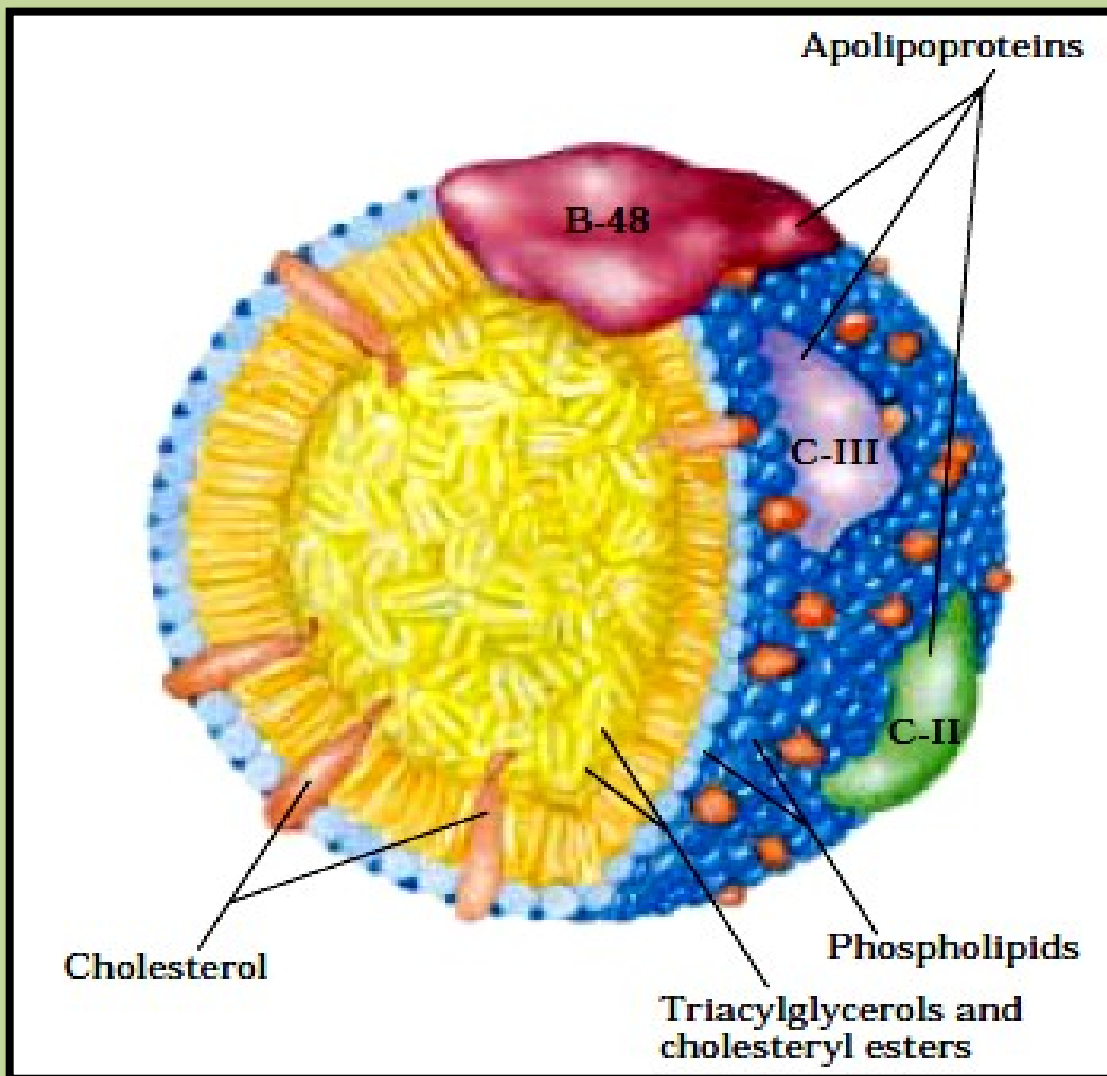


Fig-D6. Molecular Structure & Composition of a Typical Chylomicrons

Transport of Dietary Lipids in the Blood

► By the process of **exocytosis**, CMs are:

- Secreted by the intestinal epithelial cells
- Into the **chyle of the lymphatic system** &
- Enter the **blood through the thoracic duct**

► Initially, the particles are called **nascent CMs**

□ As they accept proteins from HDL within the lymph & the blood;

- they become “**mature**” CMs

Transport of Dietary...(Cont'd)

▶ HDL transfers proteins to the **nascent chylomicrons**,

- Particularly apoE & apoCII

👉 [Fig-D7](#)

□ **ApoE is recognized by membrane receptors,**

- Particularly those on the **surface of liver cells**
- Allowing **ApoE-bearing lipoproteins** to enter these cells by;
 - Endocytosis for subsequent digestion by **lysosomes**

□ **ApoCII acts as an activator of Lipoprotein lipase (LPL):**

- Primarily **within muscle & adipose tissue**
- That **digests the TAGs** of the CMs & VLDL in the blood

Blood

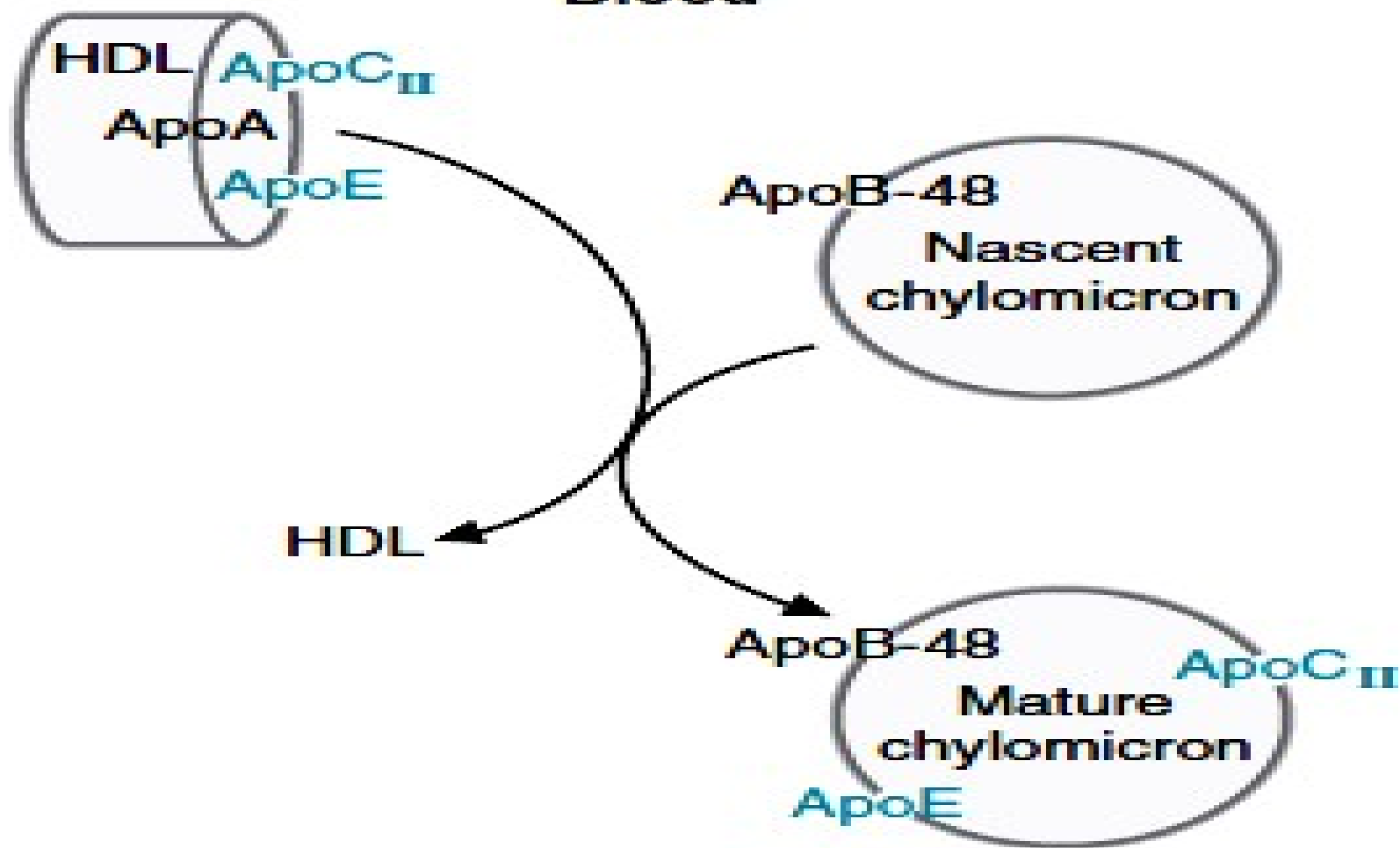


Fig-D7. Transfer of proteins from HDL to chylomicrons.

Fate of Chylomicrons

► The TAGs of the CMs are digested by  Fig-D8

□ LPL:

- Attached to the **proteoglycans in the basement membranes of endothelial cells** that line the capillary walls
- Produced by:
 - ✓ adipose cells,
 - ✓ muscle cells (particularly cardiac muscle), &
 - ✓ cells of the lactating mammary gland

□ The isozyme synthesized in adipose cells has

- **a higher K_m** than the isozyme synthesized in **muscle cells**

❖ Adipose LPL is more active after a meal

- when CMs levels are elevated in the blood

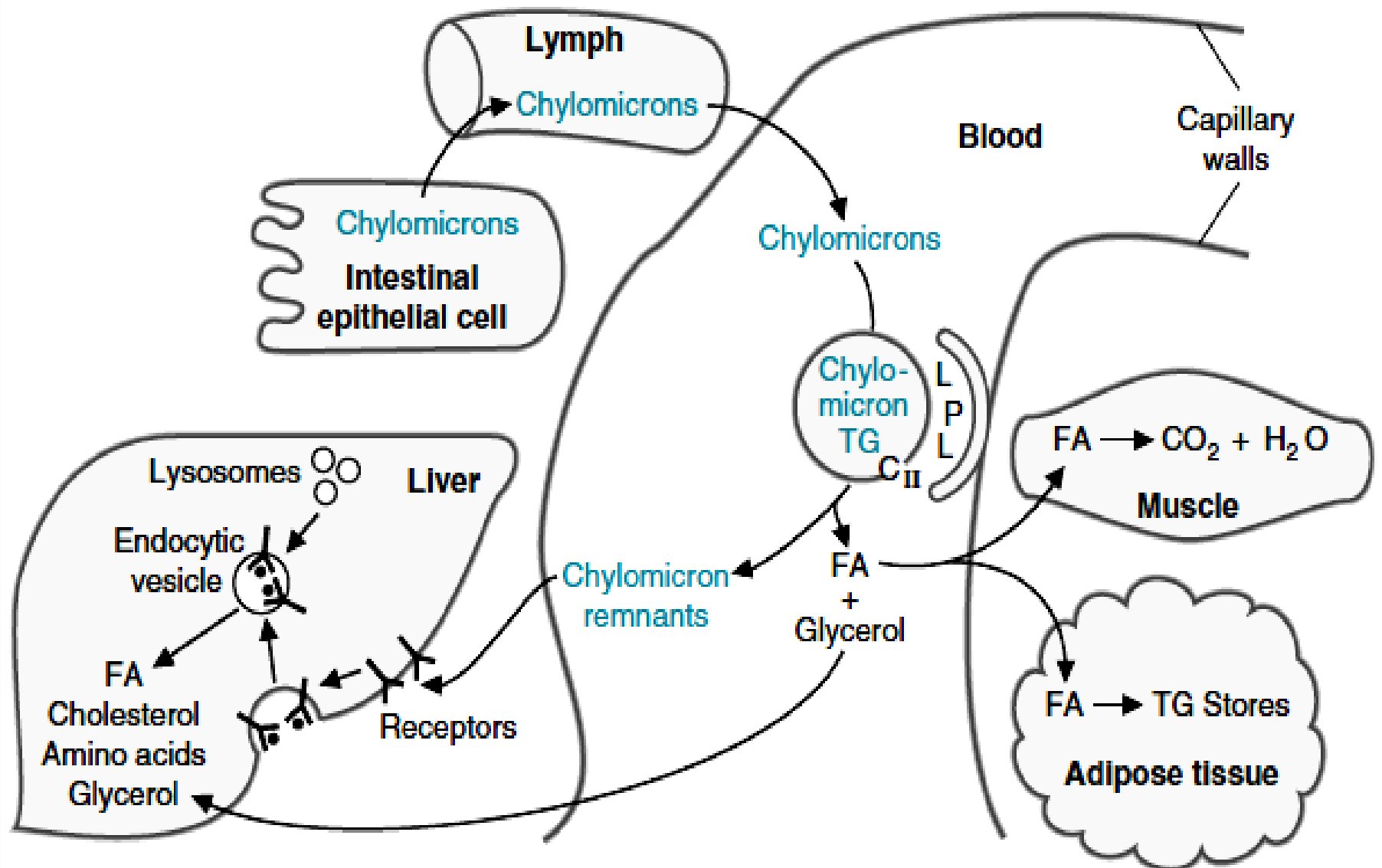


Fig-D8. Fate of Chylomicrons

Fate of...(Cont'd)

▶ Insulin stimulates:

- the synthesis & secretion of **adipose LPL**

□ after a meal, when **TAG levels increase in blood**

- LPL has been **upregulated** (through insulin release)
- to facilitate the hydrolysis of FAs from the TAG

▶ The FAs released from TAGs by LPL are

- not very soluble in water
- They become soluble in blood by
 - forming complexes with the **protein albumin**

Fate of...(Cont'd)

▶ The major fate of the FAs is

- **storage as TAG** in adipose tissue.

□ However, these FAs also may be

- **oxidized for energy in muscle & other tissues**

▶ The LPL in the capillaries of muscle cells has

- **a lower K_m than adipose LPL**

□ Thus, muscle cells can obtain **FAs from blood lipoproteins**

- whenever they are needed for energy
- even if the conc. of the **lipoproteins is low**

▶ The **glycerol** released from CM TAGs by LPL

- used for **TAG synthesis** in the liver in the fed state

Fate of...(Cont'd)

- ▶ The portion of a CM remains in the blood
 - after LPL action is known as a **chylomicron remnant**
- ▶ This remnant binds to **receptors on hepatocytes**
 - recognize apoprotein E
 - taken up by the **process of endocytosis**
- ▶ Lysosomes fuse with the endocytic vesicles,
 - the **CM remnants** are degraded by **lysosomal enzymes**
- ▶ The products of lysosomal digestion:
 - e.g., FAs, amino acids, glycerol, CL, phosphate
 - can be reused by the cell

summary of chylomicrons metabolism

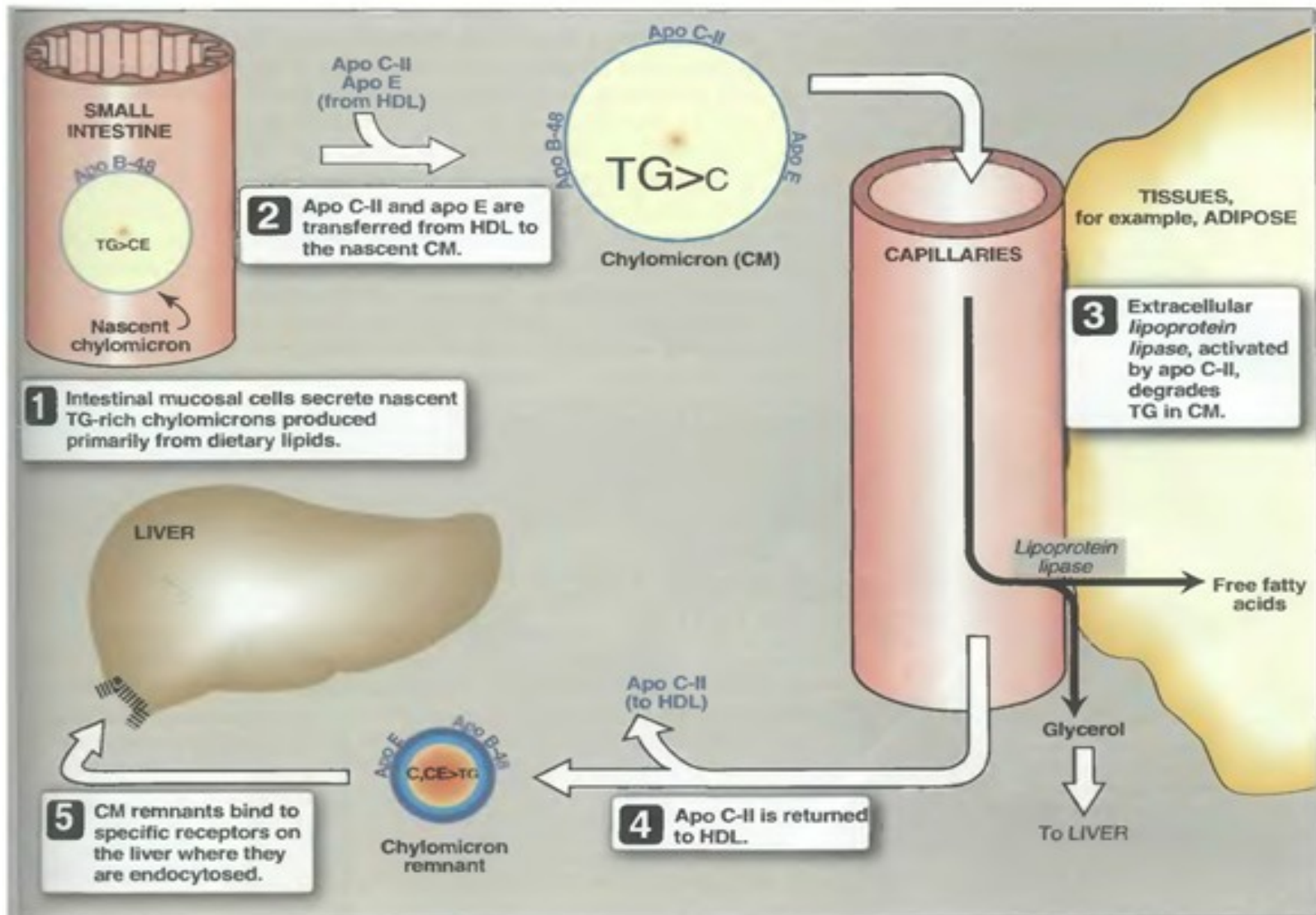


Figure 18.16

Metabolism of chylomicrons. CM = chylomicron; TG = triacylglycerol; C = cholesterol; CE = cholesteryl esters. Apo B-48, apo C-II, and apo E are apolipoproteins found as specific components of plasma lipoproteins. The lipoproteins are not drawn to scale (see Figure 18.13 for details of the size and density of lipoproteins).

Type I hyperlipidemia (familial hyperchylomicronemia)

Patients with

- deficiency of **lipoprotein lipase** or **apo C-II** show
 - » dramatic accumulation of **TAG- rich lipoproteins in the plasma**

Metabolism of very low density lipoproteins(VLDLs)

❖ VLDLs

- produced in the liver
 - composed **predominantly of TAG** and
 - carry this lipid from **the liver** to **peripheral tissues**.
- ❖ the TAG is degraded by **lipoprotein lipase** (as chylomicrons).
- ❖ **Fatty liver** occurs in conditions
- imbalance between
 - **hepatic TAG synthesis** and
 - **the secretion of VLDL**.
 - Diseases such as
 - ✓ hepatitis
 - ✓ uncontrolled DM and
 - ✓ chronic ethanol ingestion can cause **fatty liver**.

Modification of circulating VLDL

- ❖ As VLDLs pass through the circulation
 - **their structure is altered.**
- ❖ TAG is removed by LPL
 - causing the VLDL to **decrease in size** and **become more dense.**
- ❖ Surface components
 - the **apoC-II** and **apo E** are transferred to HDL.
 - **cholesteryl esters** are transferred to VLDL in an exchange reaction that
 - concomitantly transfers **TAG** or **phospholipid** to HDL.
 - This exchange is accomplished by
 - **cholesteryl ester transfer protein**

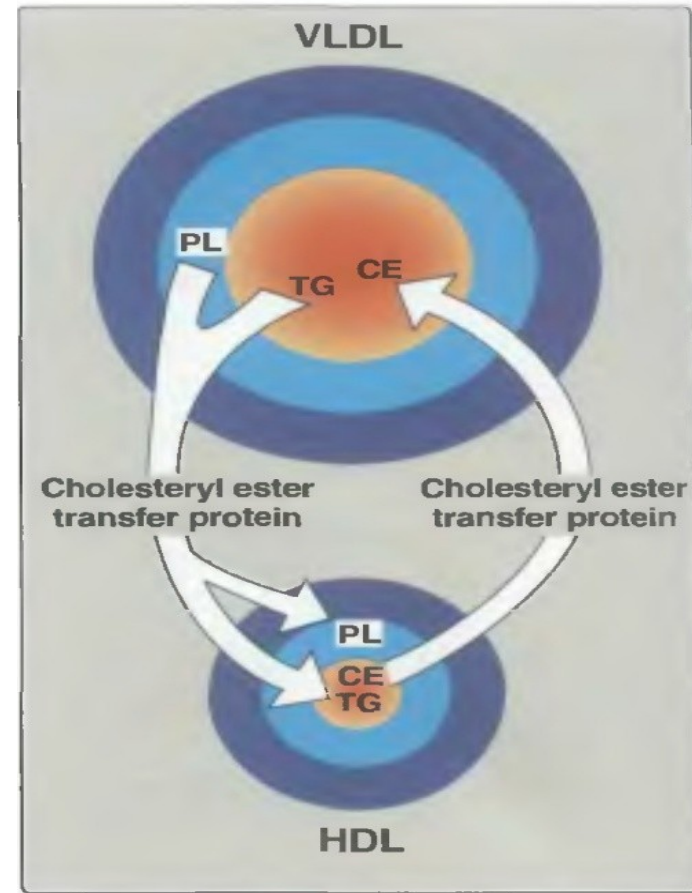


Figure 18.18

Transfer of cholesteryl esters (CE) from HDL to VLDL in exchange for triacylglycerol (TG) or phospholipids (PL).

Modification of circulating VLDL

- After these modifications
 - the **VLDL** has been converted **LDL**.
 - the (IDL) is observed during transition from VLDL to LDL
 - IDLs can also be taken up by cells through **receptor-mediated endocytosis**.

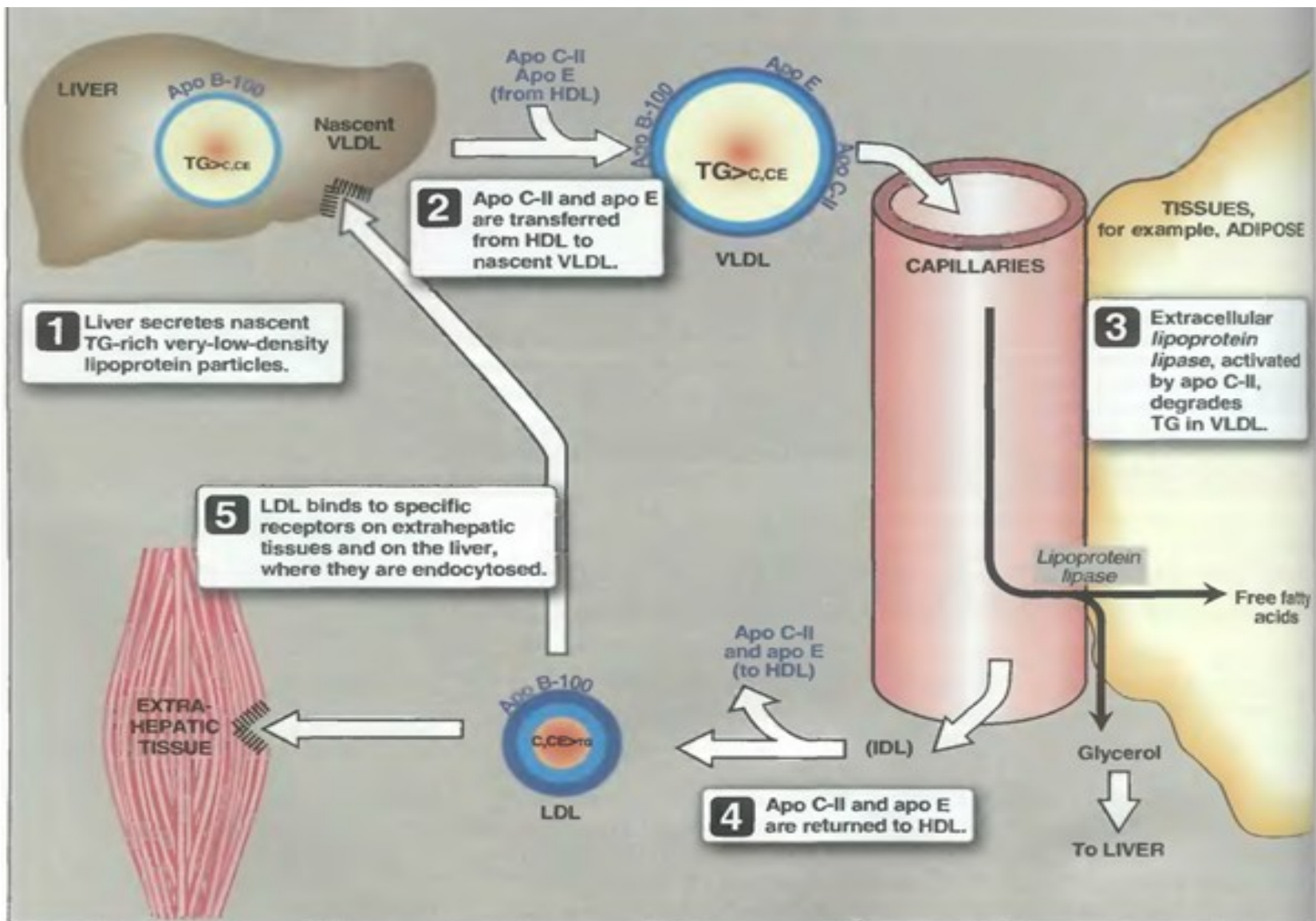


Figure 18.17

Metabolism of VLDL and LDL. TG = triacylglycerol; VLDL = very-low-density lipoprotein; LDL = low-density-lipoprotein; IDL = intermediate-density lipoprotein; C = cholesterol; CE = cholesterol esters. Apo B-100, apo C-II, and apo E are apolipoproteins found as specific components of plasma lipoproteins. Lipoproteins are not drawn to scale (see Figure 18.13 for details of the size and density of lipoproteins).

Metabolism of Low Density Lipoprotein

LDL particles

- retain **apoB – 100** , but loose their other apolipoproteins to HDL.
- contain much less TAG than their **VLDL predecessors** and
- have a high **concentration of cholesterol** and **cholesteryl esters**.

1. Receptor- mediated endocytosis:

- The primary function of LDL particles
 - to provide **cholesterol** to **peripheral tissues**.
 - by **depositing free cholesterol** on the membranes , and
 - by binding to **receptors on the cell- surface membranes** that recognize **apolipoprotein B-100**.
- LDL receptors are
 - negatively charged **glycoprotein molecules**.
- A deficiency of **functional LDL receptors** causes
 - a significant **elevation in plasma LDL** and
 - therefore of **plasma cholesterol**,

Metabolism of Low Density Lipoprotein

but plasma **TAG levels** remain normal

- **type II hyperlipidemia (familial –hyperbeta-lipoproteinemia).**
- This can greatly accelerate the progress of **atherosclerosis**.

2. After binding

- the **LDLs are internalized** by **endocytosis**.

3. The vesicle containing the LDL

- rapidly loses its **clathrin coat** and
- fuses with **other similar vesicles**
- forming **larger vesicles** called **endosomes**

Metabolism of Low Density Lipoprotein

4. The pH of the contents of the endosome falls

- due to the **proton-pumping activity** of **endosomal ATPase**
- allowing the **separation of the LDL from its receptor**.
- The **receptors** then migrate to one side of the endosome whereas,
- the **LDLs stay free** within the **lumen of the vesicle**.
 - This structure is called CURL – the **compartment for uncoupling of receptor and ligand**.

5. The **receptors can be recycled**, whereas,

- the **lipoprotein remnants** in the vesicle are degraded by **lysosomal enzymes**
 - **cholesterol**
 - **amino acids**
 - **fatty acids and**
 - **phospholipids**
- These compounds can be recycled by the cell

Metabolism of Low Density Lipoprotein

6. The **number of LDL –receptors** varies according to the
- availability of **these lipoprotein particles** and
 - **needs of the cell.**

For example

- if there is a large amount of circulating plasma LDL
 - ✓ **the number of cell- surface receptors for it decreases**
 - ✓ frequently termed “**down –regulation**”.
- Conversely, if cells **are starved for cholesterol**
 - ✓ they **increase the number of cell- surface receptors**
 - ✓ “up- regulation”.

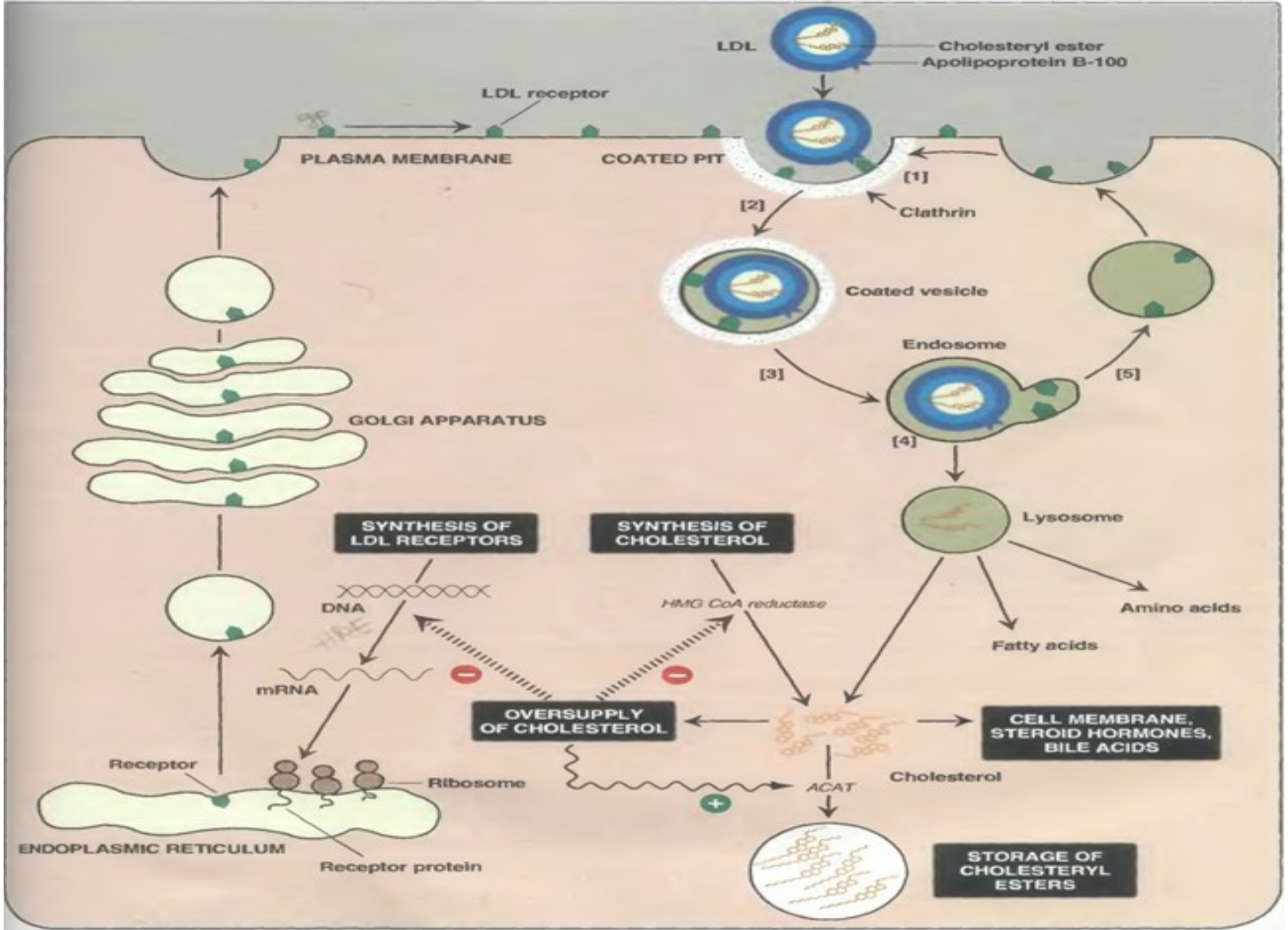


Figure 18.20

Cellular uptake and degradation of LDL. ACAT = acyl CoA:cholesterol acyltransferase.

Metabolism of high-density lipoproteins (HDLs)

HDL particles

- synthesized in the liver and
- Released into the blood stream by exocytosis.

They perform a number of important functions:

1. Serve as a circulating **reservoir of apo C-II and apo –E**
2. Removes free cholesterol from **extrahepatic tissues** and **esterifying** it
 - using the plasma enzyme **phosphatidylcholine : cholesterol acyltransferase**, (PCAT—also known as LCAT, where” L “stands for lecithin).
3. **Transfers cholesterol** to VLDL and LDL in exchange for **TAG**.
4. **Carries CE** to the liver, where
 - the HDL is degraded and cholesterol released.

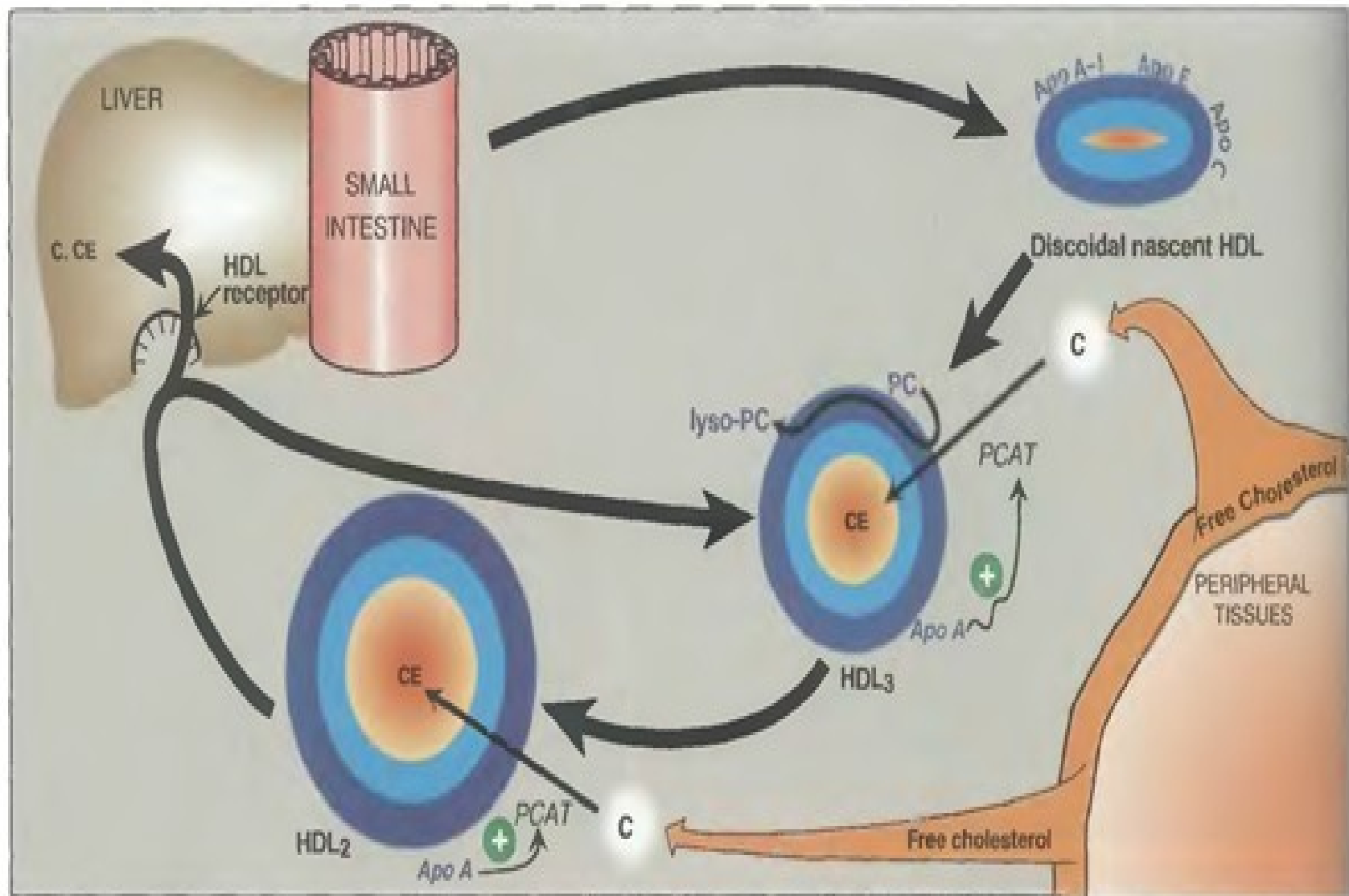


Figure 18.23

Metabolism of HDL. PC = phosphatidylcholine; lyso-PC = lysophosphatidylcholine. PCAT = *Phosphatidylcholine cholesterol transferase*.

HDL uptake of free cholesterol

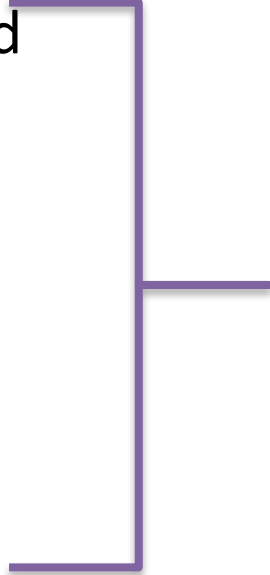
- ❖ Newly secreted HDLs are **disc shaped particles**
 - containing predominantly
 - **un-esterified cholesterol,**
 - **phospholipid** (largely phosphatidylcholine) and
 - **a number of apolipoproteins**
 - ✓ apoE, apoA-I and apoC-II.
- ❖ They are rapidly converted to spherical particles
 - ❖ as **they accumulate cholesterol.**
- ❖ HDL particles are
 - excellent acceptors of unesterified cholesterol from
 - the **surface of cell membranes** and
 - **from other circulating lipoproteins.**

Esterification of free cholesterol

- ❖ Once free cholesterol is taken up by the HDL particle,
 - it is immediately esterified by PCAT,
 - a plasma enzyme synthesized by the liver
 - ✓ which is activated by **apoA-1** of the HDL.
 - Plasma levels of **apoA-1 are increased** by **modest alcohol intake**.
 - The fatty acid from carbon 2 of phosphatidylcholine
 - transferred directly to **cholesterol** leaving **lysophosphatidylcholine**.
- ❖ Spherical HDL particles are taken up by
 - the liver by **receptor-mediated endocytosis**, and
 - **cholesteryl esters** are degraded.
 - **The cholesterol** released can be
 - ✓ either repackaged in lipoproteins
 - ✓ converted into bile acids or secreted into the bile for removal from the body.

Reverse cholesterol transport

❖ The selective transfer of cholesterol from

- **peripheral cells** to HDLs and
 - from HDLs to the liver for
 - bile acids synthesis or
 - disposal via the bile and
 - to **steroidogenic cells** for
 - **hormone synthesis is**
- 

➤ This is in part the basis for

- the inverse relationship seen between **plasma HDL concentration** and **atherosclerosis** and for HDL's designation as the “**good cholesterol carrier**”.

Disorder of Plasma Lipoproteins

Disorder of Plasma Lipoproteins

- Inherited disorders of lipoproteins are encountered in some individuals resulting in **primary hyper- or hypolipoproteinemias**. These are due to
 - genetic defects in lipoprotein metabolism and transport.
- The **secondary acquired** lipoprotein disorders are due to
 - some other diseases, resulting in abnormal lipoprotein pattern which often resembles the primary inherited condition
 - e.g. diabetes mellitus, nephrotic syndrome, atherosclerosis hypothyroidism etc.

INTRODUCTION

- ❖ Hyperlipidemia is defined as abnormally raised/elevated levels of any or all lipids in the blood.
- ❖ It results from abnormalities in lipid metabolism or plasma lipid transporters or a disorder in the synthesis and degradation of plasma lipoproteins.
- ❖ Hyperlipidemia is a broad term also called hyperlipoproteinemia is a common disorder in developed countries.
- ❖ It is the most common form of dyslipidemia.
- ❖ Lipoproteins are transported in a protein capsule where the size of the capsule or lipoprotein determines its density. Lipoproteins of low density are the main cause for hyperlipidemia.

CLASSIFICATION

- Hyperlipidemia may be basically classified as:
 - I. **Familial (Primary hyperlipidemia)**: Caused by specific genetic abnormalities
 - II. **Secondary hyperlipidemia** : Results from another underlying disorder that leads to alterations in the plasma lipid and protein metabolism.
 - III. **Idiopathic hyperlipidemia**: Cause is unknown.

FAMILIAL HYPERLIPIDEMIA

➤ Familial hyperlipidemia are classified according to the Fredrickson classification, which is based on the pattern of lipoproteins on electrophoresis or ultracentrifugation . It does not account for HDL directly and it does not distinguish among the different genes that may be partially responsible for some of these conditions. It can be classified as:

- I. Hyperlipoproteinemia type 1 (Familial lipoprotein lipase deficiency)
- II. Hyperlipoproteinemia type 2 (FH & PH)
- III. Hyperlipoproteinemia type 3 (Familial dysbetalipoproteinemia)
- IV. Hyperlipoproteinemia type 4 (Hypertriglyceridemia)
- V. Hyperlipoproteinemia type 5 (Familial combined hyperlipidemia)

Familial Hyperlipidemia

Fredrickson classification of hyperlipidemias			
Hyperlipo-proteinemia		Defect	Increased lipoprotein
Type I	a	Decreased lipoprotein lipase (LPL)	Chylomicron
	b	Altered ApoC2	
	c	LPL inhibitor in blood	
Type II	a	LDL receptor deficiency	LDL
	b	Decreased LDL receptor and increased ApoB	LDL and VLDL
Type III		Defect in Apo E 2 synthesis	IDL
Type IV		Increased VLDL production and decreased elimination	VLDL
Type V		Increased VLDL production and decreased LPL	VLDL & Chylomicron

❑ Hyperlipidemias are also classified as in which type of lipids are elevated:

✓ HYPERCHOLESTEROLEMIA

✓ HYPERTRIGLYCERIDEMIA or

✓ BOTH IN CASE OF COMBINED
HYPERLIPIDEMIA

Hyperlipoproteinemias (Hyperlipidemia)

- **Type I:** This is due to **familial lipoprotein Lipase** deficiency.
 - The enzyme defect causes increase in **plasma chylomicron** and **triacylglycerol levels**.
- **Type IIa:** This is also known as **hyperbetalipoproteinemia** and is caused by a defect in **LDL receptors**.
- **Secondary type IIa hyperlipoproteinemia** is observed in association with diabetes mellitus, hypothyroidism, nephrotic syndrome etc.
- This disorder is characterized by **hypercholesterolemia**.

Hyperlipoproteinemias (Hyperlipidemia)

- **Type IIb** : Both LDL and VLDL increase along with elevation in plasma cholesterol and triacylglycerol.
 - This is believed to be due to overproduction of **apoB**.
- **Type III** : This is commonly known as broad beta disease and characterized by the appearance of a **broad β -band corresponding to intermediate density lipoprotein (IDL)** on electrophoresis
- **Type IV**: This is due to overproduction of endogenous triacylglycerols with a concomitant rise in VLDL.
 - Type IV disorder is usually associated with **obesity, alcoholism, diabetes mellitus** etc.
- **Type V**: Both chylomicrons and VLDL are elevated.
 - This is mostly a secondary condition, due to disorders such as obesity, diabetes and excessive alcohol consumption etc

Hypolipoproteinemias

- Although low levels of plasma lipids (not HDL) within the normal range may be beneficial to the body, very low lipid levels are undesirable.
- These are commonly associated with certain abnormalities
- **1.Familial hypobetalipoproteinemia:** is an inherited disorder probably due to an **impairment in the synthesis of apoprotein B** .
- The plasma LDL concentration in the affected individuals is between 10 to 50% of normal values.
- This disorder is harmless, and, the individuals have healthy and long life.

Hypolipoproteinemias

2. Abetalipoproteinemia:

- This is a rare disorder due to a defect in the synthesis of apoprotein B.
- It is characterized by a total absence of β -lipoprotein (LDL) in plasma.
- Triacylglycerols are not found in plasma, but they accumulate in liver and intestine.
- Serum cholesterol level is low
- Associated with decreased absorption of fat and fat – soluble vitamins.
- Impairment in physical growth and mental retardation are commonly observed.

Hypolipoproteinemias

Familial alpha-lipoprotein deficiency (Tangier disease) :

- The plasma HDL particles are almost absent.
- Due to this, the reverse transport of cholesterol is severely affected leading to the accumulation of cholesteryl esters in tissues.
- An absence of apoprotein CII-which activates lipoprotein lipase-is also found.
- The plasma triacylglycerol levels are elevated.
- The affected individuals are at an increased risk for atherosclerosis

Summary of the Hyperlipidemia

TABLE 14.6 Classification and characteristics of hyperlipoproteinemias (hyperlipidemias)

<i>Hyperlipoproteinemia Type</i>	<i>Increased plasma lipoprotein(s)</i>	<i>Increased plasma lipid (most)</i>	<i>Probable metabolic defect</i>	<i>Risk of atherosclerosis</i>	<i>Suggested treatment</i>
I	Chylomicrons	Triacylglycerols	Deficiency of lipoprotein lipase	May increase	Low fat diet
IIa	LDL	Cholesterol	Deficiency of LDL receptors	Very high (mostly in coronary artery)	Low cholesterol fat diet; cholestyramine
IIb	LDL and VLDL	Triacylglycerols and cholesterol	Overproduction of apo-B	— do —	— do —
III	IDL	Triacylglycerols and cholesterol	Abnormality in apo-E	Very high (mostly in peripheral vessels)	Low fat and low caloric diet; clofibrate
IV	VLDL	Triacylglycerols	Overproduction of TG	May or may not increase	Low fat and low caloric diet; niacin
V	Chylomicrons and VLDL	Triacylglycerols	—	— do —	— do —

Classification of Total, LDL, and HDL Cholesterol, and Triglycerides

Total cholesterol

<200	Desirable
200–239	Borderline high
≥240	High

LDL cholesterol

<100	Optimal
100–129	Near or above optimal
130–159	Borderline high
160–189	High
≥190	Very high

HDL cholesterol

<40	Low
≥60 mg/dL	High

Triglycerides

<150	Normal
150–199	Borderline high
200–499	High
≥500	Very high

All values are given in milligrams per deciliters.

Physical signs of Hyperlipidaemia

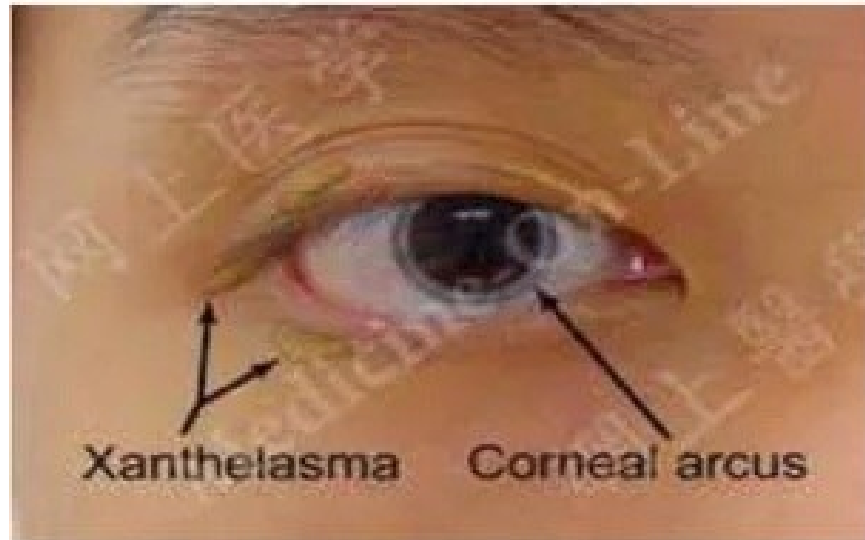
Xanthomata Tuberosa



Xanthelasma Palpebrarum



Corneal arcus



Lipemia retinalis

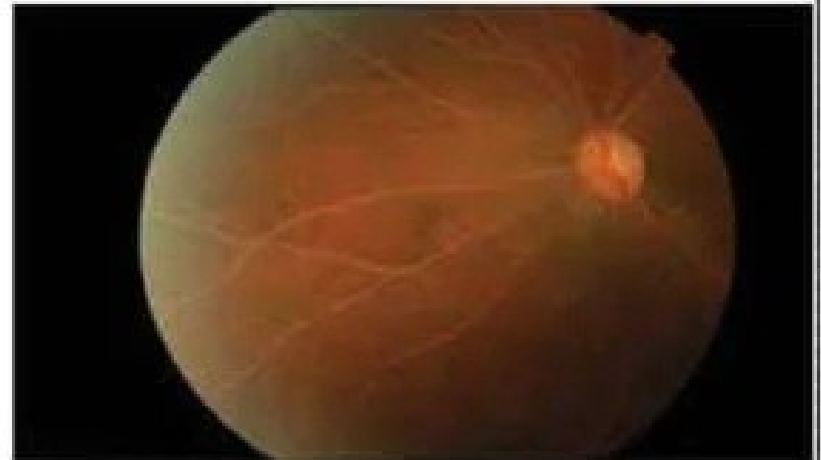


Fig. 1: Milky white appearance of retinal veins and arteries seen suggestive of lipemia retinalis

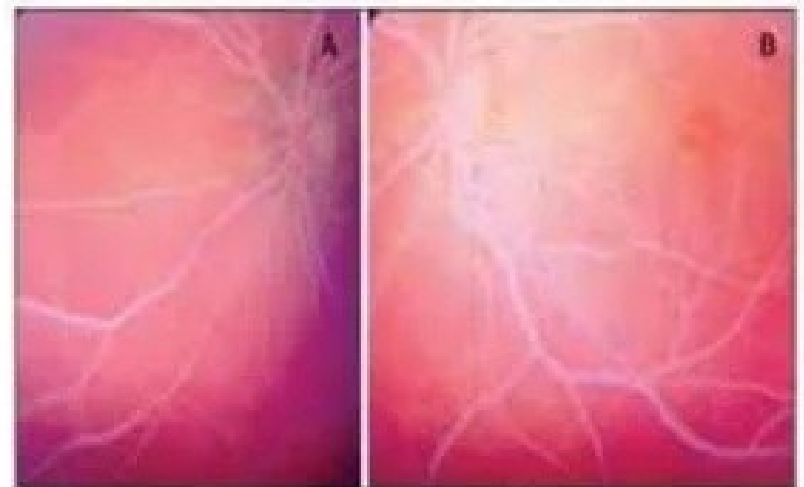


Figure 1 - A right eye and B left eye both with lipemia retinalis grade III: all vessels with a creamy appearance, difficult making it to differentiate retinal arterioles from veins

GENERAL APPROACH TO TREATMENT

Establishing targeted changes and outcomes with consistent reinforcement of goals and measures at follow-up visits to attain goals are important to reduce barriers for optimizing Therapeutic Lifestyle Changes(TLC) and pharmacologic therapy. TLC should be implemented in all patients prior to considering drug therapy. The components of TLC include reduced intake of saturated fats and cholesterol, dietary options to reduce LDL, such as consumption of plant stanols and sterols and soluble fibre, weight reduction, and increased physical activity.

In general, physical activity of moderate intensity 30 minutes per day for most days of the week should be encouraged. Patients with known CAD or who are at high risk should be evaluated before they undertake vigorous exercise. Weight and BMI should be determined at each visit, and lifestyle patterns to induce a weight loss of 10% should be discussed with persons who are overweight. All patients should be counselled to stop smoking.

NON-PHARMACOLOGIC TREATMENT

DIET

Eat a heart-healthy diet

Making changes to your diet can lower your “bad” cholesterol levels and increase your “good” cholesterol levels. Here are a few changes you can make:

Choose healthy fats. Avoid saturated fats that are found primarily in red meat, bacon, sausage, and full-fat dairy products. Choose lean proteins like chicken, turkey, and fish when possible. Switch to low-fat or fat-free dairy. And use monounsaturated fats like olive and canola oil for cooking.

Cut out the trans fats. Trans fats are found in fried food and processed foods, like cookies, crackers, and other snacks. Check the ingredients on product labels. Skip any product that lists “partially hydrogenated oil.”

Eat more omega-3s. Omega-3 fatty acids have many heart benefits. You can find them in some types of fish, including salmon, mackerel, and herring. They can also be found in some nuts and seeds, like walnuts and flax seeds.

Increase your fiber intake. All fiber is heart-healthy, but soluble fiber, which is found in oats, brain, fruits, beans, and vegetables, can lower your LDL cholesterol levels.

Eat more fruits and veggies. They're high in fiber and vitamins and low in saturated fat.

Weight

Being overweight is a risk factor for hyperlipidemia and heart disease. Losing weight can help reduce LDL, total cholesterol, and triglyceride levels. It can also boost HDL, which helps to remove the bad cholesterol out of the blood.

To lose weight, adopt a low-calorie diet and increase your physical activity

Physical activity

A lack of physical activity is a risk factor for heart disease.

Regular exercise and activity helps lower LDL, raise HDL, and encourage weight loss. At least 30 minutes of physical activity is recommended, at least 5 days a week. Brisk walking is an excellent and easy choice.

Quit smoking

Smoking triggers many problems that contribute to heart disease. It promotes plaque buildup on the walls of the arteries, increases LDL levels, and it encourages blood clot formation and inflammation. Quitting smoking will result in higher HDL. This may be one reason why cardiovascular disease risk falls after quitting.

Macronutrient Recommendations for the Therapeutic Lifestyle Changes Diet

Component ^a	Recommended Intake
Total fat	25%–35% of total calories
Saturated fat	Less than 7% of total calories
Polyunsaturated fat	Up to 10% of total calories
Monounsaturated fat	Up to 20% of total calories
Carbohydrates ^b	50%–60% of total calories
Cholesterol	<200 mg/day
Dietary fibre	20–30 g/day
Plant sterols	2 g /day
Protein	Approximately 15% of total calories
Total calories	To achieve and maintain desirable body weight

^a Calories from alcohol not included.

^b Carbohydrates should derive from foods rich in complex carbohydrates, such as whole grains, fruits, and vegetables.

ANTIHYPERLIPIDEMIC DRUGS

HMG CoA REDUCTASE INHIBITORS

- *Atorvastatin*
- *Fluvastatin*
- *Lovastatin*
- *Pravastatin*
- *Rosuvastatin*
- *Simvastatin*

FIBRATES

- *Fenofibrate*
- *Gemfibrozil*

NIACIN

BILE ACID SEQUESTRANTS

- *Colesevelam*
- *Colestipol*
- *Cholestyramine*

CHOLESTEROL ABSORPTION INHIBITORS

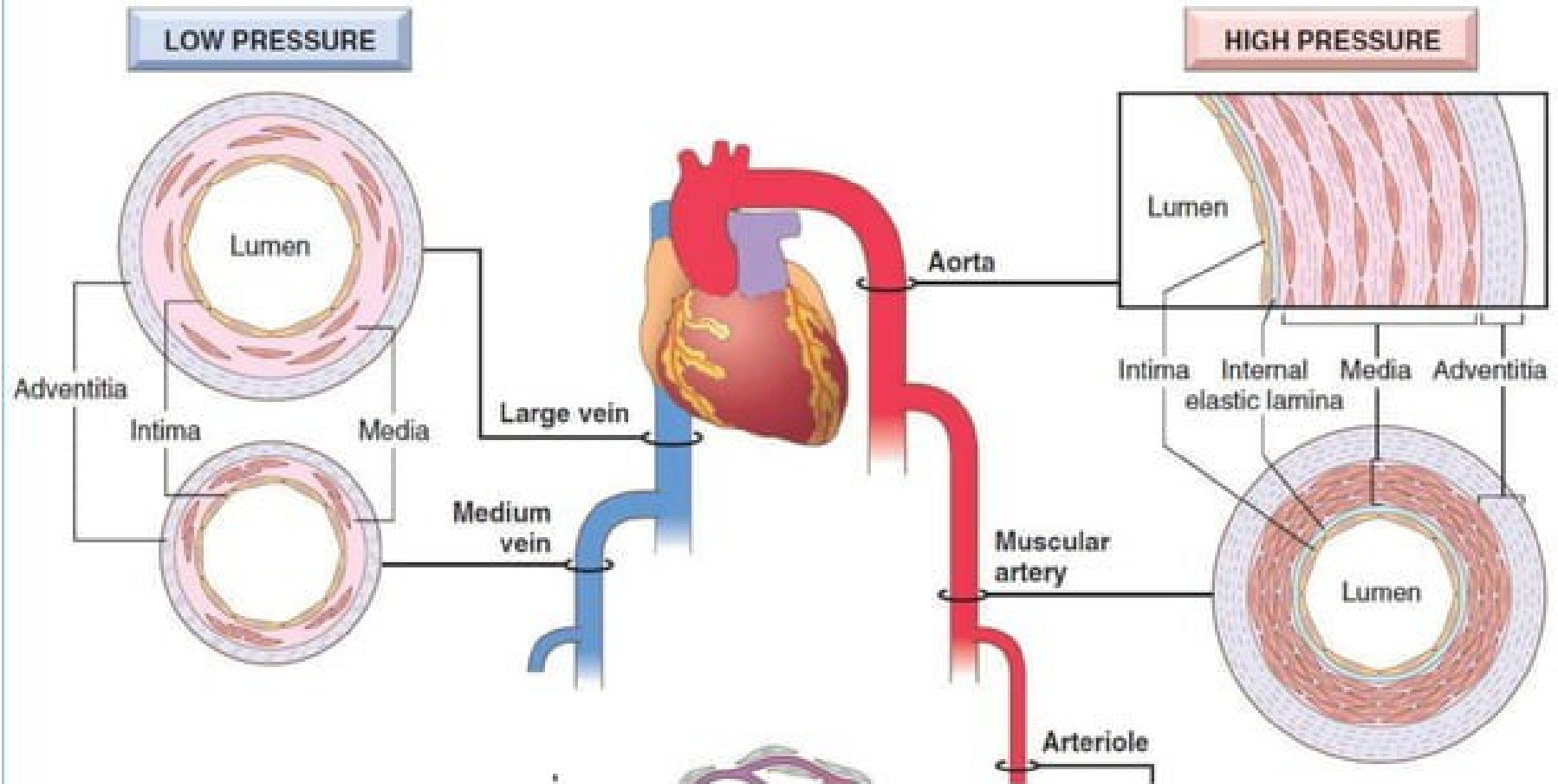
- *Ezetimibe*

Atherosclerosis

Structure of blood vessels

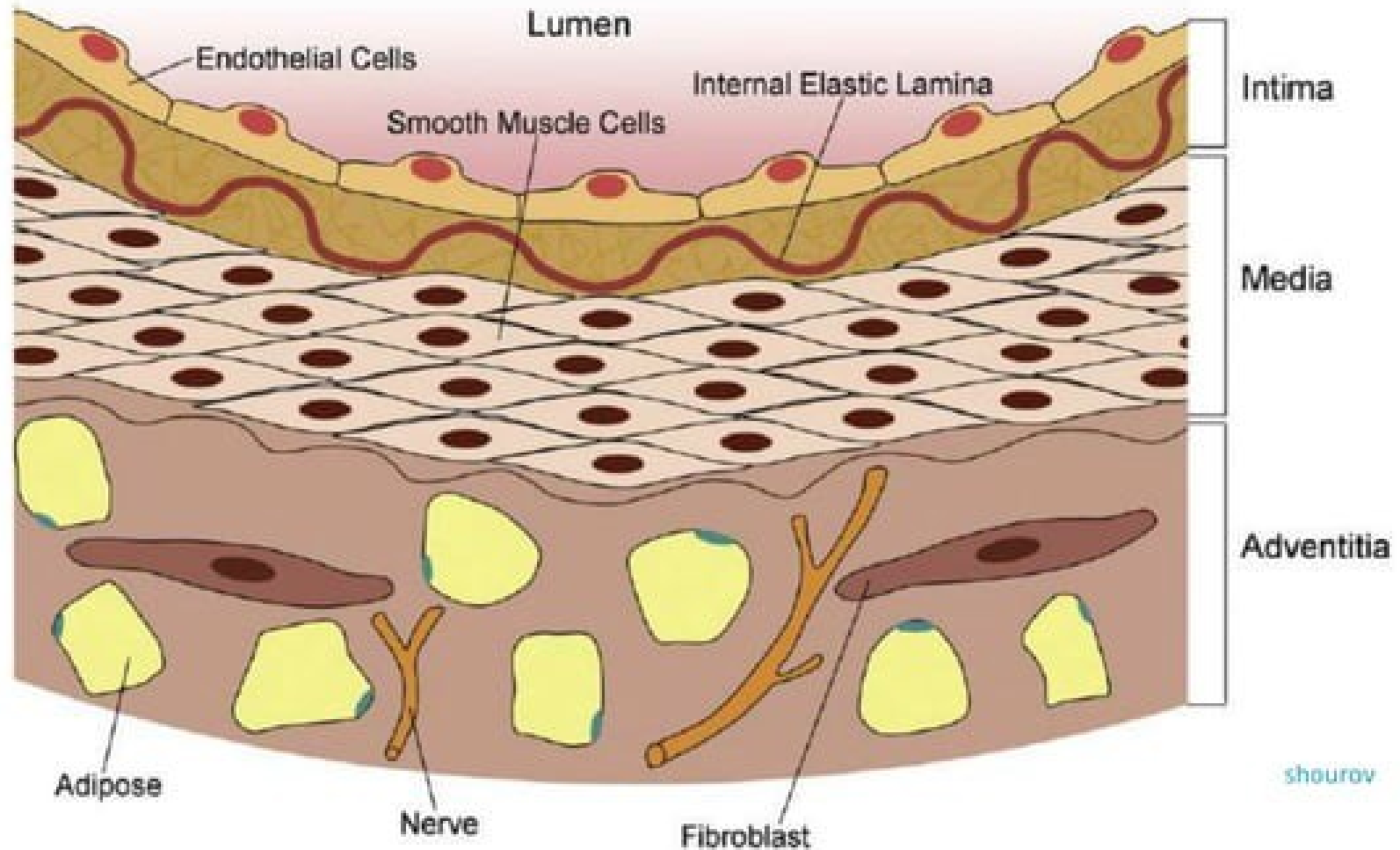
LOW PRESSURE

HIGH PRESSURE



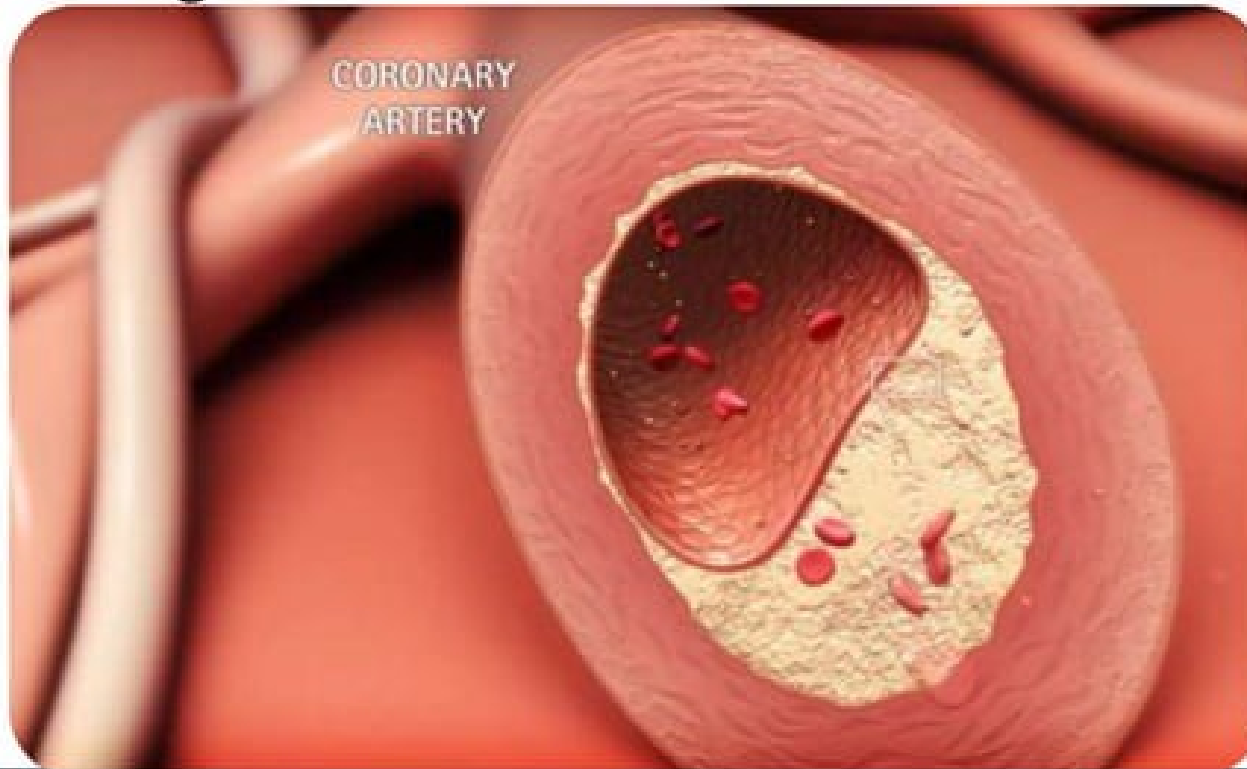
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Vessel walls are organized into three concentric layers:
intima, media, and adventitia

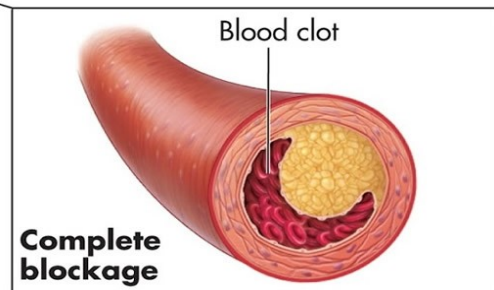
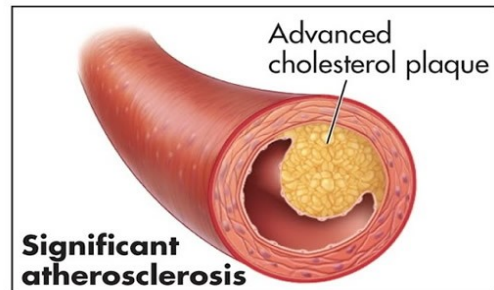
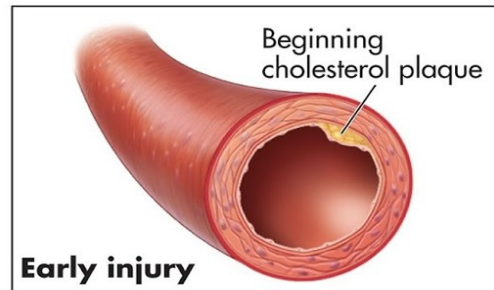
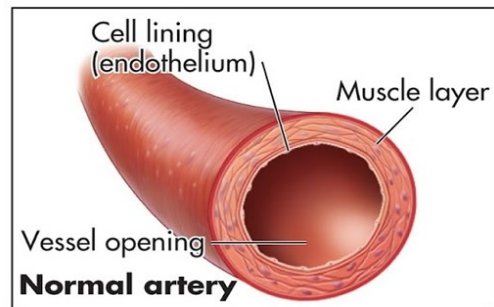
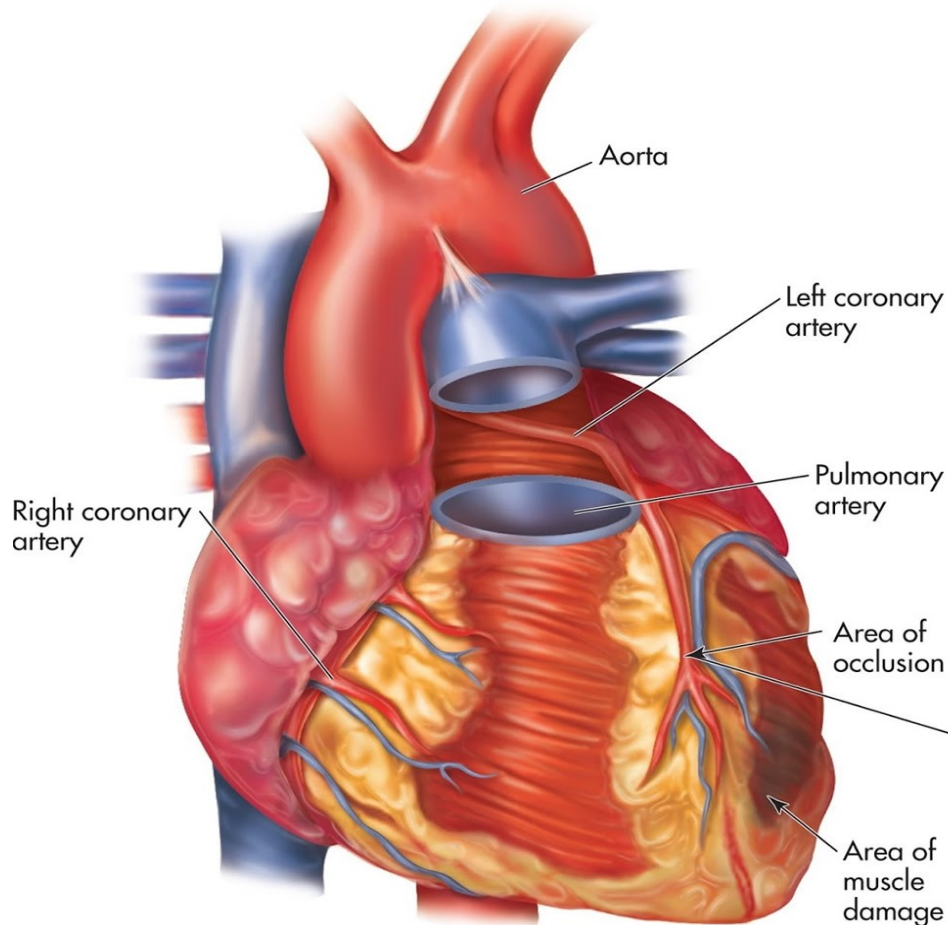


Atherosclerosis

- Atherosclerosis= atheros (gruel/paste) + sclerosis (hardness),
- Arteriosclerosis literally means “hardening of the arteries”;
- the formation of fibrofatty lesions in the intimal lining of the large and medium-size arteries



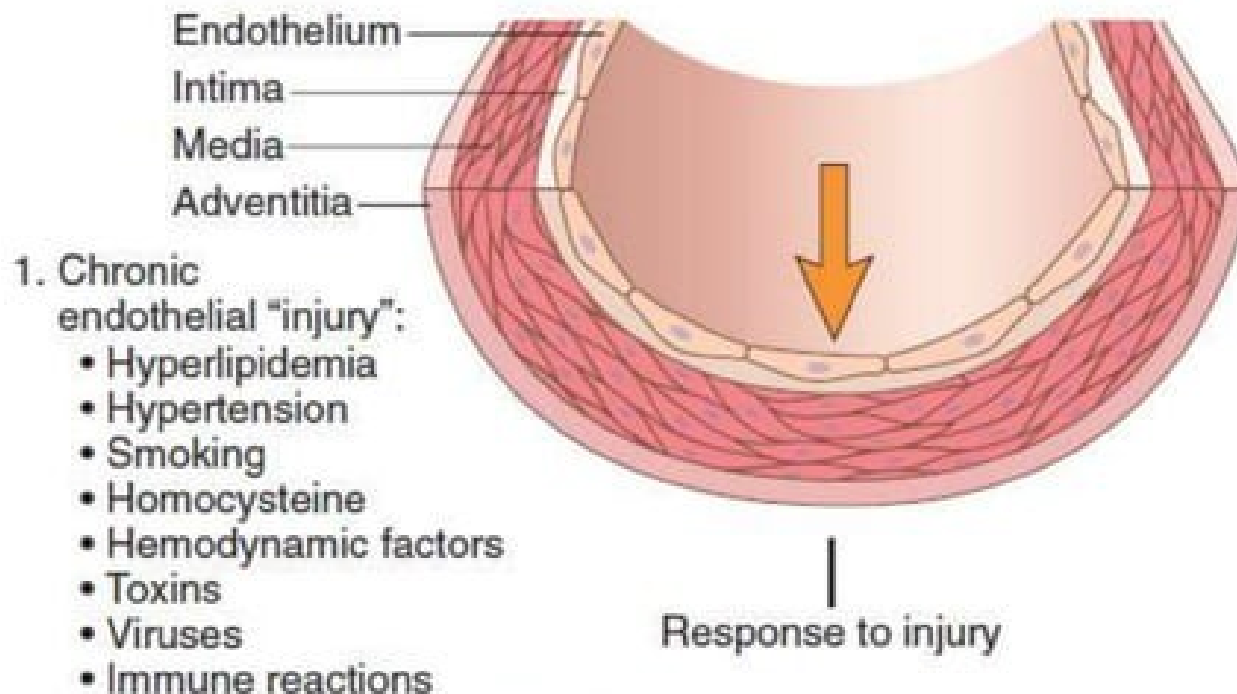
Atherosclerosis



Pathogenesis of Atherosclerosis

Step 01: Endothelial dysfunction/ Endothelial injury

- increased permeability, leukocyte adhesion, and thrombosis.
- which stimulates the accumulation and oxidation of LDL in the vessel wall



Pathogenesis of Atherosclerosis

Step 02: Accumulation of lipoproteins

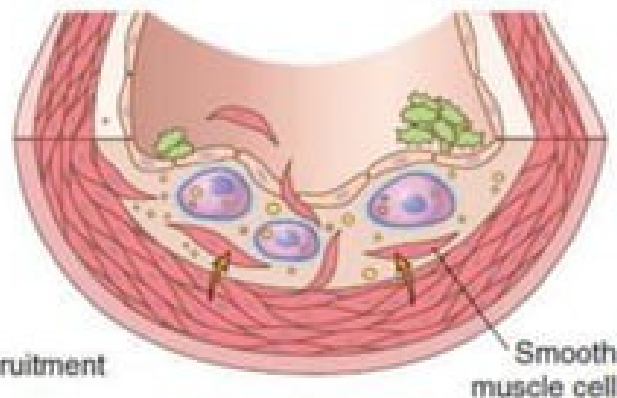
mainly oxidized LDL and cholesterol crystals in the vessel wall

2. Endothelial dysfunction (e.g., increased permeability, leukocyte adhesion), monocyte adhesion and emigration



Step 03: Platelet adhesion

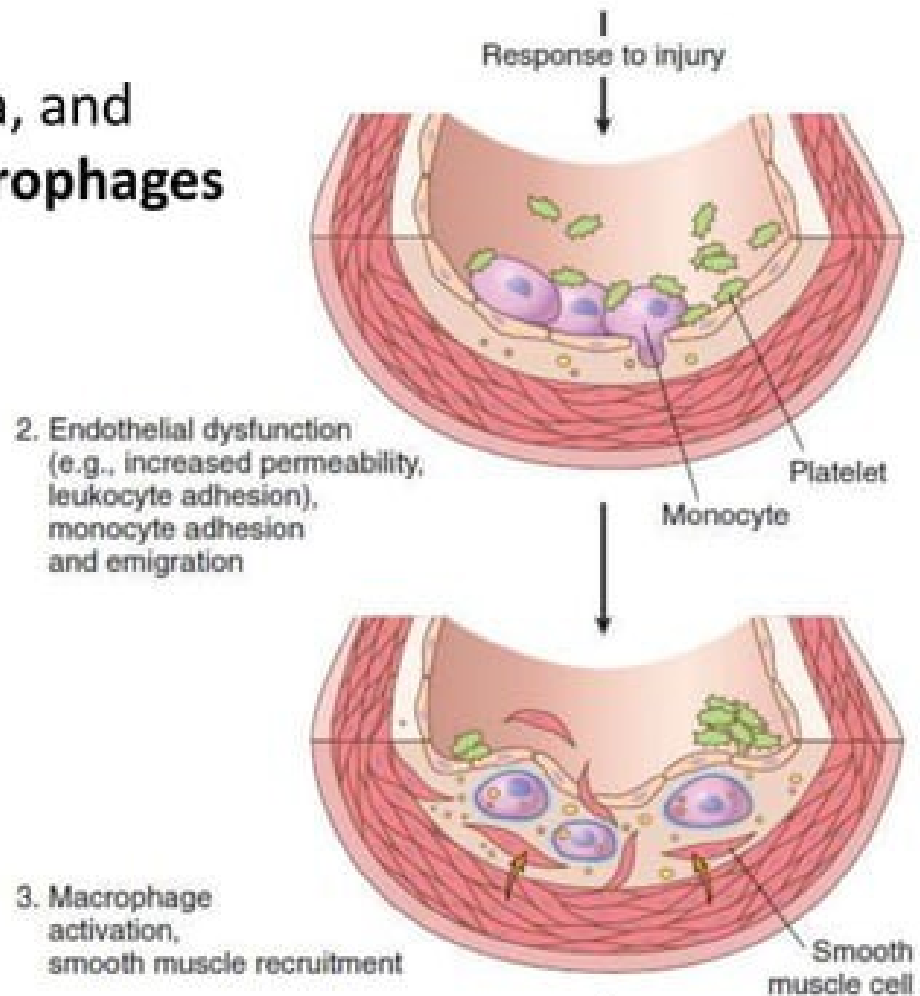
3. Macrophage activation, smooth muscle recruitment



Pathogenesis of Atherosclerosis

Step 04: Monocyte adhesion to the endothelium

Migration into the intima, and differentiation into **macrophages** and **foam cells**



Pathogenesis of Atherosclerosis

Step 05: Lipid accumulation

within macrophages, which release inflammatory cytokines

Step 06: Smooth muscle cell recruitment

due to factors released from activated platelets, macrophages, and vascular wall cells

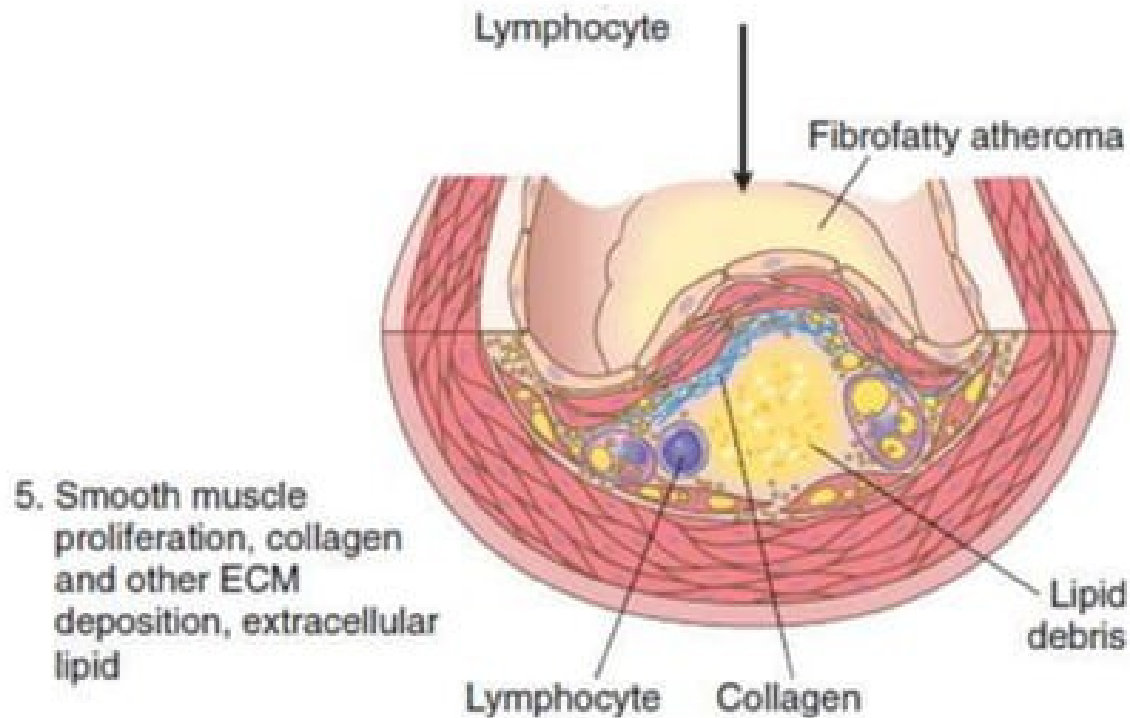
4. Macrophages and smooth muscle cells engulf lipid



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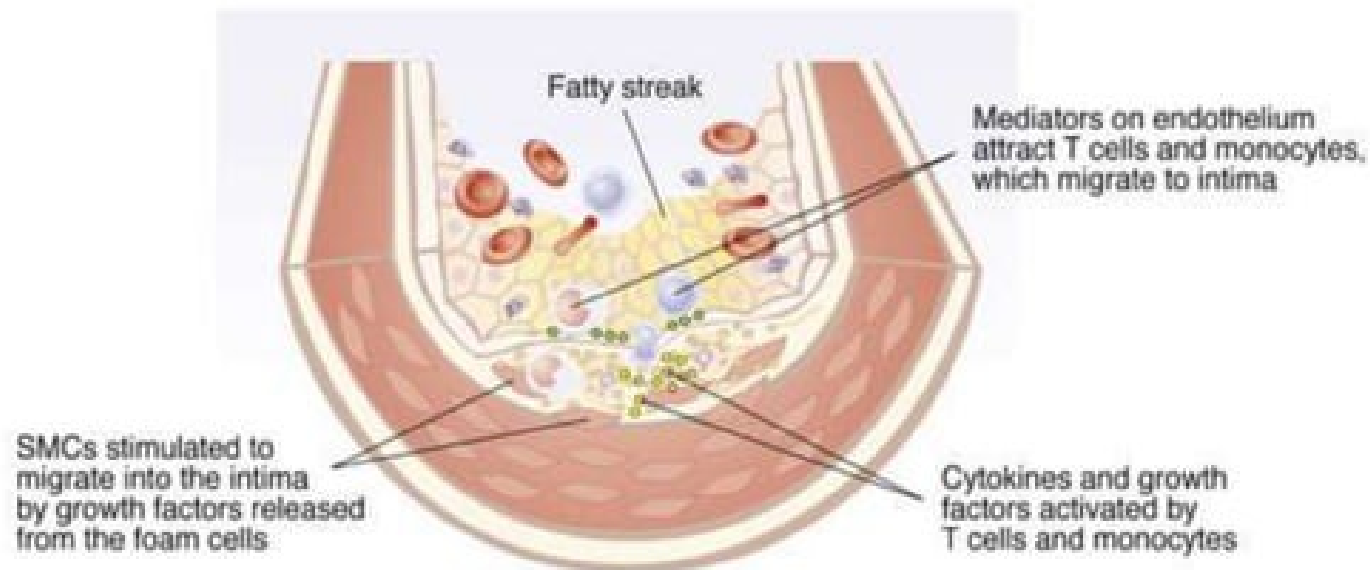
Pathogenesis of Atherosclerosis

Step 07: Smooth muscle cell proliferation and ECM production



Fatty Streaks

Fatty streaks begin as minute yellow, flat macules that coalesce into elongated lesions. They are composed of lipid-filled foamy macrophages



Atherosclerotic Plaque

have three principal components:

- ❖ **Cells** (smooth muscle cells, macrophages & T cells)
- ❖ **Extracellular matrix** (collagen, elastic fibers & proteoglycans)
- ❖ **Intracellular and extracellular lipid**

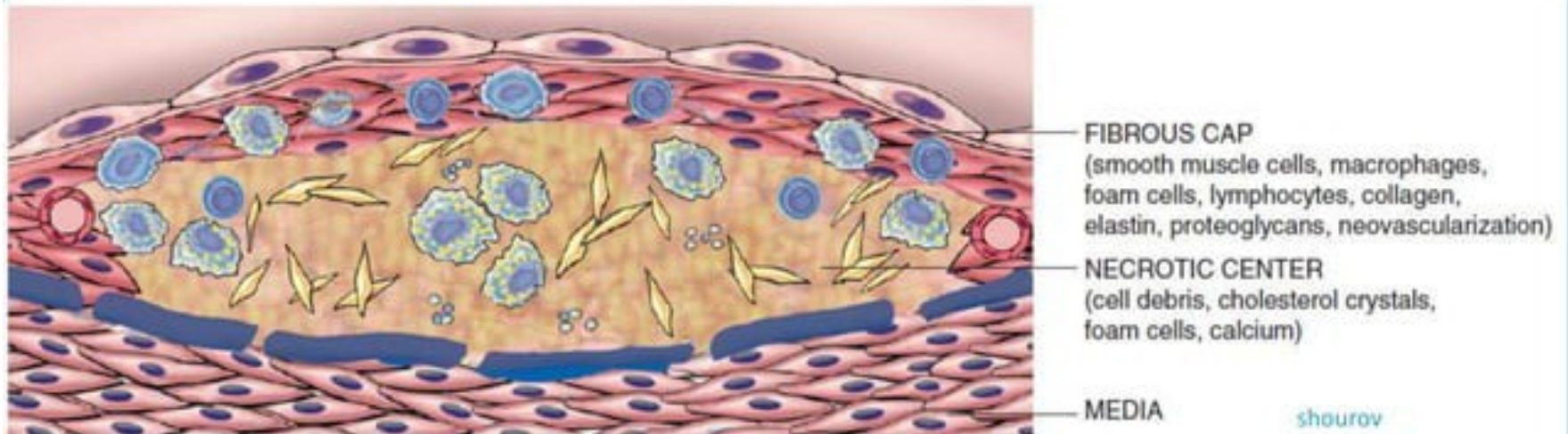
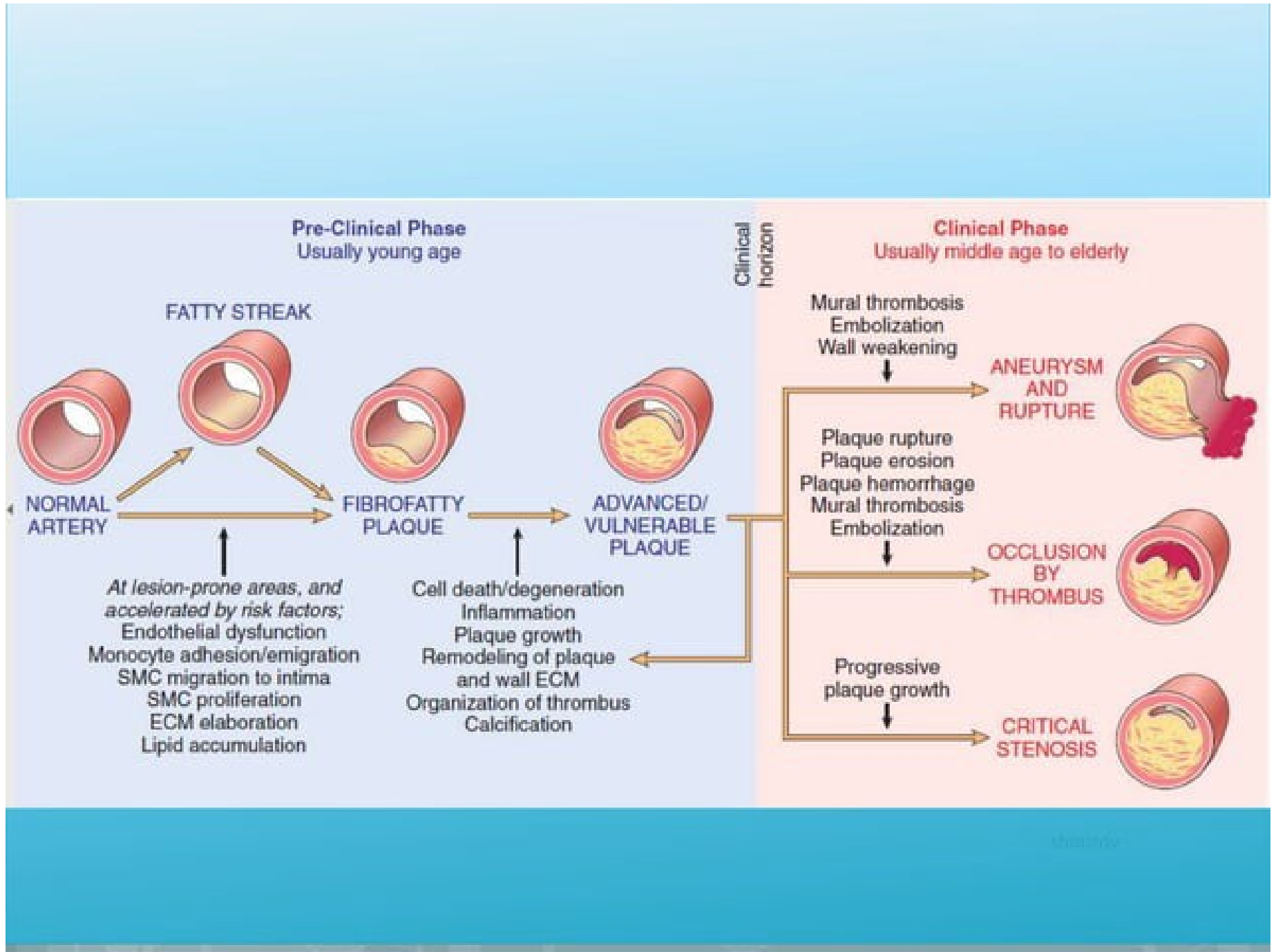
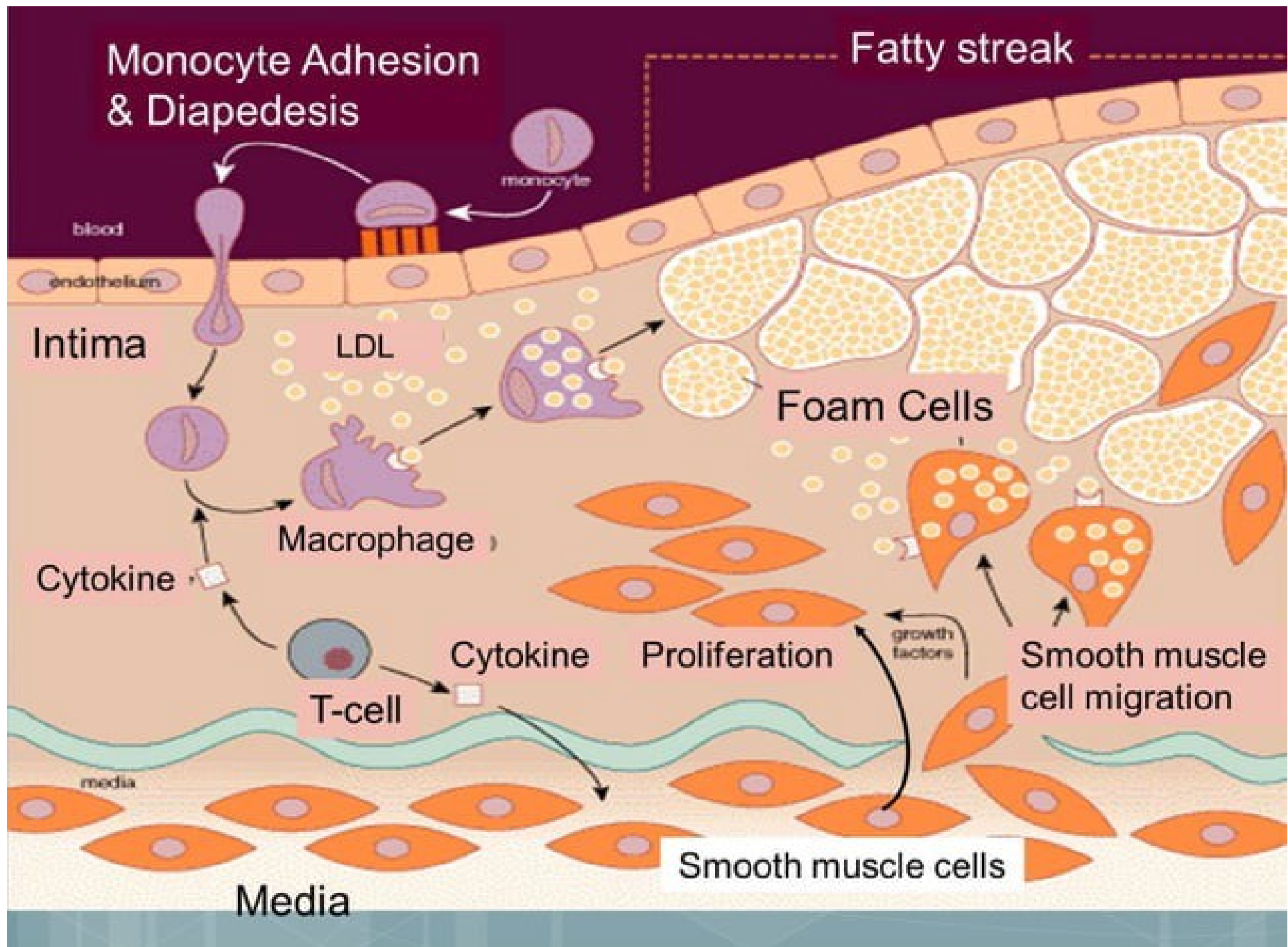


Figure 9-7 The basic structure of an atheromatous plaque.

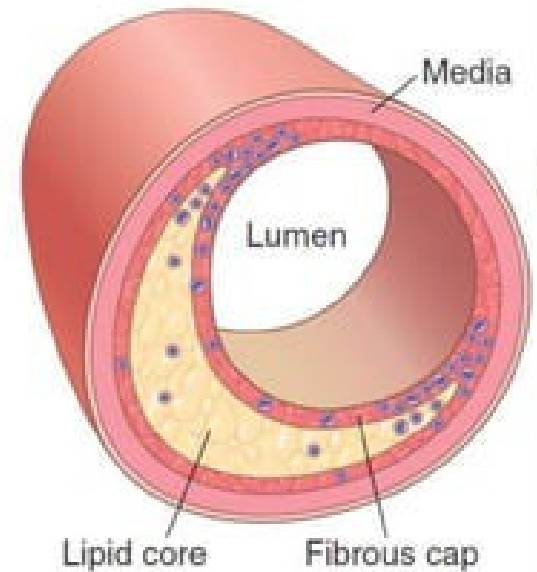




Plaques

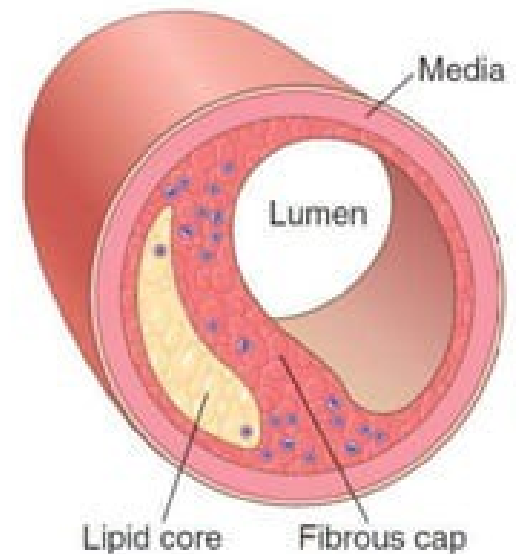
Vulnerable plaques have

- thin fibrous caps,
- large lipid cores
- increased inflammation.



Stable plaques have

- densely collagenized
- thickened fibrous caps
- minimal inflammation
- negligible underlying atheromatous cores



Thank you!!!